

SolarLab Technical and User Manual

Technical guide to thin-film solar-cell simulation, workflows, and validation

SolarLab Project

2026-08-11

Contents

1	Executive Summary	5
2	How To Read This Manual	6
3	Simulator Architecture	7
3.1	Solver And Model Scope	8
4	What SolarLab Simulates	9
5	Beginner Physics Primer	12
5.1	Electrons, Holes, And Bands	12
5.2	Generation And Recombination	12
5.3	Mobile Ions And Hysteresis	12
6	Device Definition	14
6.1	Core Data Flow	14
6.2	Layer Roles	14
6.3	Main Material Parameters	14
6.4	Parameter Dictionary	15
6.4.1	Electrical And Transport Fields	15
6.4.2	Ion And Hysteresis Fields	16
6.4.3	Recombination And Trap Fields	17
6.4.4	Optical Fields	17
6.4.5	Temperature And Field-Dependent Mobility Fields	17
6.4.6	Device-Level Fields	18
6.4.7	Advanced Model Options	19
6.5	Global Device Parameters	20
6.6	Built-In Potential	20
6.7	YAML Example	22
7	Governing Equations	24
7.1	Electrostatics: The Poisson Equation	24
7.2	Carrier Transport: Continuity And Drift-Diffusion	25
7.2.1	Heterojunction Band Alignment	26
7.2.2	Effective-Density-of-States Correction	26
7.3	Heterointerface Thermionic Emission And Tunnelling	27
7.3.1	Thermionic-Field-Emission Tunnelling	28
7.4	Ion Migration	29
7.4.1	Legacy Whole-Flux Steric Form	30

7.5	Recombination	30
7.5.1	Shockley-Read-Hall (Trap-Assisted) Recombination	31
7.5.2	Radiative Recombination And Photon Recycling	31
7.5.3	Auger Recombination	32
7.5.4	Interface Recombination	32
7.6	Optical Generation	34
7.6.1	Beer-Lambert Absorption	34
7.6.2	Transfer-Matrix Optics	35
7.7	Field-Dependent Mobility	36
7.8	Temperature Dependence	36
7.9	Spatial Trap Profiles	37
7.10	Band-Gap Grading	37
8	Boundary And Initial Conditions	39
8.1	Potential	39
8.2	Carrier Contacts	39
8.3	Ion Boundaries	40
8.4	Initial States	40
9	Physics Tiers	41
10	Numerical Method	43
10.1	Spatial Grid	44
10.2	Method Of Lines And State Vector	44
10.3	Scharfetter-Gummel Flux Discretization	44
10.4	Poisson Fast Path	45
10.5	Right-Hand-Side Assembly	46
10.6	Time Integration And Robustness	46
10.7	Direct Steady-State Solver	47
10.8	Cancellation-Safe Quasi-Fermi Steady State	47
10.9	Operator Splitting For Slow Ion Dynamics	48
10.10	Terminal Current Evaluation	49
10.11	Two-Dimensional Extension	50
10.12	Numerical Safety	50
11	Running SolarLab	51
11.1	Installation	51
11.2	Backend	51
11.3	Frontend	51
11.4	Python API	51
11.5	Worked Example 1: Baseline J-V Sweep	52
11.6	Worked Example 2: Absorber Thickness Study	52
11.7	Worked Example 3: Enabling TMM Optics	53
12	Backend API	54
13	Experiment Manual	56
13.1	J-V Sweep	56
13.2	Current Decomposition	57
13.3	Spatial Profiles	57

13.4	Dark J-V	58
13.5	Suns- V_{oc}	58
13.6	$V_{oc}(T)$	58
13.7	EQE / IPCE	59
13.8	Electroluminescence	59
13.9	Impedance	60
13.10	Mott-Schottky	60
13.11	Transient Photovoltage	61
13.12	Degradation	61
13.13	Tandem J-V	61
13.14	2D J-V	62
13.15	$V_{oc}(L_g)$ Grain Sweep	62
14	Shipped Presets	63
15	Troubleshooting And Diagnostics	64
15.1	voc_bracketed=false	64
15.2	Sentinel Or Zero FF/PCE	64
15.3	TMM Or EQE Fails Because Optical Data Are Missing	64
15.4	YAML Values Look Numeric But Behave Strangely	65
15.5	Slow Or Stiff Runs	65
15.6	Unphysical Band Barriers	65
15.7	2D Results Differ From 1D	65
15.8	Frontend Job Appears To Hang	65
16	Validation And Evidence	66
16.1	Numerical Evidence And Model-Scope Figures	68
16.1.1	Other Regression Contracts	71
16.2	Validation Traceability Matrix	72
16.3	SCAPS Comparison: Model Alignment Table	73
17	Limitations And Best Practices	77
17.1	Model Assumptions	77
17.2	Known Formulation Limitations	77
17.3	Data And Optical Limits	78
17.4	Numerical Limits	78
17.5	2D Limits	79
17.6	Reporting Checklist	79
17.7	Before Publication	79
18	SolarScale Screening Bridge	80
19	References	81
20	Appendix A: Documentation Sources	83
21	Appendix B: Figure Sources	85
22	Appendix C: Glossary	86

List of Figures

3.1	SolarLab architecture and data flow	7
4.1	Electrical coordinate, layer order, and contact-potential sources	10
4.2	Band bending, quasi-Fermi levels, and abrupt-interface closures	11
10.1	Numerical drivers, variable sets, and certification paths	43
16.1	Registered c-Si QF J-V and C-V grid-ladder observations	69
16.2	Physical-interface CBO response, grid contraction, and certification gates	70
16.3	Registered 1D/2D parity domain and current model scope	71

Chapter 1

Executive Summary

SolarLab is a thin-film photovoltaic device simulator that couples drift-diffusion transport, Poisson electrostatics, mobile-ion redistribution, recombination, and optical generation within a reproducible software workflow. The numerical core is exposed through a Python API, a FastAPI service layer, and a TypeScript/Plotly workstation. An experimental two-dimensional extension supports lateral microstructure and grain-boundary studies.

This edition covers the cancellation-safe quasi-Fermi steady-state solver, its small-signal frequency-domain extension, and the optional zero-thickness heterointerface model. At that interface, the quasi-Fermi solver removes the ordinary Scharfetter-Gummel face and solves thermionic transport and interface-SRH recombination together. Unsupported physics produces an error before Newton starts. The transient and direct steady-state solvers are still available for the device classes they support.

New device files also state how the contact voltage is obtained. SolarLab can calculate the signed built-in potential from the two semiconductor contact work functions or from two specified metal work functions. Carrier exchange at the contacts is configured separately. A manual V_{bi} is reserved for legacy benchmarks in which that value forms part of the published comparison.

This manual serves two purposes:

1. It introduces the device variables, parameter conventions, and simulation workflow for readers who are new to SolarLab or to semiconductor device simulation.
2. It documents the implemented physics, numerical method, validation gates, assumptions, and limitations needed for technical review and reproducible use.

Chapter [16](#) lists the test runs, benchmark ranges and source state used for this edition. The following chapter collects the model assumptions and known limits.

Chapter 2

How To Read This Manual

If you are new to device simulation, read the chapters in order through *Device Definition*, *Governing Equations*, and *Numerical Method*. These chapters explain the device, the variables, the equations, and the numerical method before introducing the software workflow.

If you are already familiar with semiconductor drift-diffusion, start with:

- Chapter 7 (*Governing Equations*), which defines the physical model in continuous form;
- Chapter 10 (*Numerical Method*), which describes the discretization, the time integrator, and the steady-state solver;
- Chapter 11 (*Running SolarLab*), which covers the YAML/Python/API workflows;
- Chapter 16 (*Validation And Evidence*), which gives validation evidence and limitations.

The notation in this manual follows standard solar-cell convention:

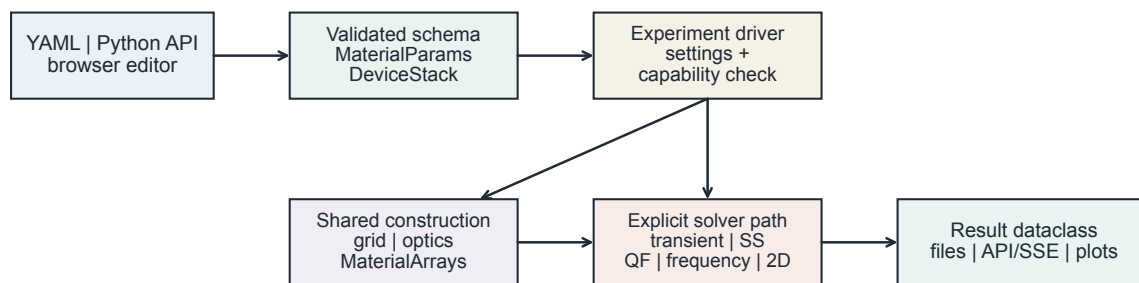
- V_{oc} : open-circuit voltage.
- J_{sc} : short-circuit current density.
- FF : fill factor.
- PCE : power-conversion efficiency.
- n : electron density.
- p : hole density.
- P : mobile positive ion-vacancy density.
- ϕ : electrostatic potential.
- E : electric field.
- G : optical generation rate.
- R : recombination rate.

All simulator inputs are SI unless explicitly stated otherwise. Length is in meters, time in seconds, current density in $A m^{-2}$, density in m^{-3} , mobility in $m^2 V^{-1} s^{-1}$, and diffusion coefficient in $m^2 s^{-1}$. The electron affinity χ and band gap E_g are stored in eV.

Chapter 3

Simulator Architecture

SolarLab uses one device schema throughout the program. YAML files, the Python API, the FastAPI backend and the TypeScript frontend all create the same `DeviceStack` object, so a value edited in the browser reaches the same solver input as a value loaded from YAML.



No implicit solver substitution. Unsupported physics stops before the numerical solve.

Figure 3.1: SolarLab architecture and data flow

Four implementation rules keep these layers consistent:

- device definition is separated from experiment settings;
- material and grid arrays are cached before the time integrator runs;
- physics hooks are activated by the resolved simulation mode and by the presence of required parameters;
- backend and frontend layers transport result dataclasses instead of reimplementing solver logic.

The experiment selects its solver explicitly. `transient`, `steady_state` and `quasi_fermi` use different state variables and support different physics. SolarLab never swaps one for another automatically; an unsupported combination stops before the solve.

3.1 Solver And Model Scope

No single driver includes every optional model. The physics tier chooses which configured terms may run, while the experiment driver sets the unknowns, boundary conditions and numerical checks. Record both with the result.

Driver	Intended quantity	Physical model	Main limitation
transient	state-preserving J-V, hysteresis and time response	density variables; coupled electrons, holes and configured mobile ions; full-tier Robin contacts, field mobility and non-local recycling where configured	the final state depends on the voltage/time history and settling protocol
steady_state	ion-frozen DC comparison	direct density-log Newton on the established 1D RHS; the seed ion profile is held fixed	ions do not relax; deep-CBO points may need the validated transient fallback
quasi_fermi	cancellation-safe local DC J-V	quasi-Fermi increments, eliminated Poisson response and separate continuity/current checks; optional zero-thickness interface model	limited to ion-free local physics with ideal carrier pins; unsupported hooks stop before Newton
quasi_fermi_frequency	restricted small-signal C-V/impedance	linear response about a certified local QF operating point	limited to the local model supported by the QF operating point
1D operator split	long-time ion/degradation evolution	ions advance with carriers frozen, followed by electronic re-equilibration	first-order split; check macro-step halving against the coupled transient
2D	lateral parity and microstructure	2D electrostatics and carrier transport with the configured static ion background	ions remain fixed during the 2D J-V solve; validation covers the registered parity and microstructure cases

full is a feature setting, not an accuracy grade. It enables the widest set of hooks available to the selected driver, but support and validation still depend on the device and experiment. For each reported curve, record the device stack, physics tier, experiment driver and solver settings.

Chapter 4

What SolarLab Simulates

SolarLab represents a solar cell as a sequence of material layers between two contacts. The default coordinate follows the stack direction. Incident light generates electron-hole pairs, while the electrostatic field and contact selectivity drive carrier separation and extraction. In perovskite devices, mobile ionic defects can redistribute under the same field, producing history-dependent current-voltage response.

The default solver is one-dimensional and is appropriate for:

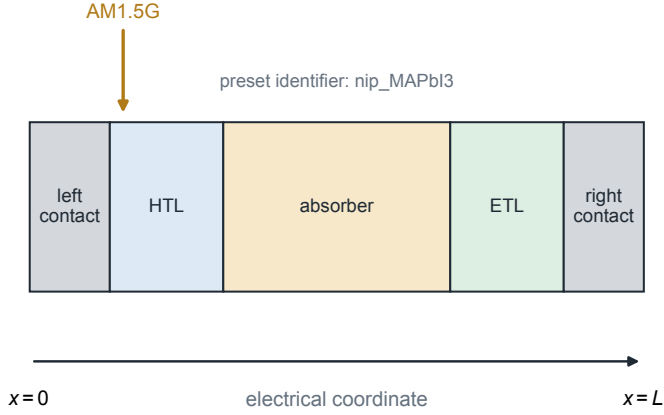
- J-V sweeps and hysteresis;
- dark diode curves;
- EQE and electroluminescence;
- impedance;
- residual-certified local QF J-V and frequency-domain C-V for supported ion-free stacks;
- opt-in abrupt-interface transport/recombination studies with a zero-thickness physical boundary;
- transient photovoltage;
- ion-coupled degradation;
- tandem current matching;
- screening many material candidates.

SolarLab also includes an experimental two-dimensional solver. The 2D solver extrudes the same vertical device stack laterally and is intended for:

- 1D/2D parity checks;
- vertical grain boundaries;
- lateral microstructure;
- $V_{oc}(L_g)$ grain-size sweeps.

The 2D solver is not a replacement for the 1D workflow. Mobile ions are held as a static Poisson background during 2D J-V runs; this is an explicit model assumption and should be reported when interpreting 2D results.

(a) Electrical coordinate and shipped layer order



(b) Electrostatic source and carrier exchange

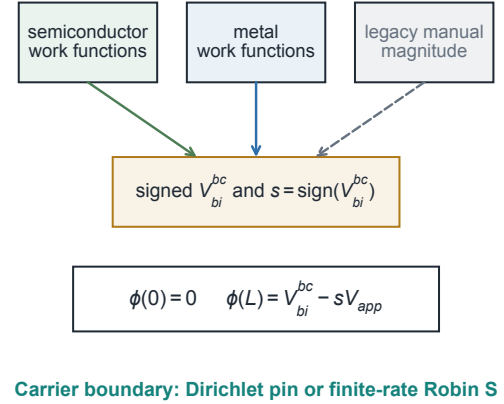


Figure 4.1: Electrical coordinate, layer order, and contact-potential sources

Illumination is always incident at $x = 0$, through the first listed electrical layer after any optical-only substrate. The shipped nip_MAPbI3 identifier therefore refers to a stored preset, not to a rule for which transport layer faces the light. Its YAML order is HTL, absorber, ETL. When porting a structure from the literature, check the layer sequence and the $x = 0$ convention rather than inferring the orientation from the preset name.

The Poisson contact potential and the carrier contact law are separate inputs. New physical configurations obtain the signed contact potential from endpoint semiconductor work functions or from two explicit metal work functions. The manual magnitude remains only for legacy comparisons. Carrier exchange is then set independently by Dirichlet pins or finite-rate Robin velocities.

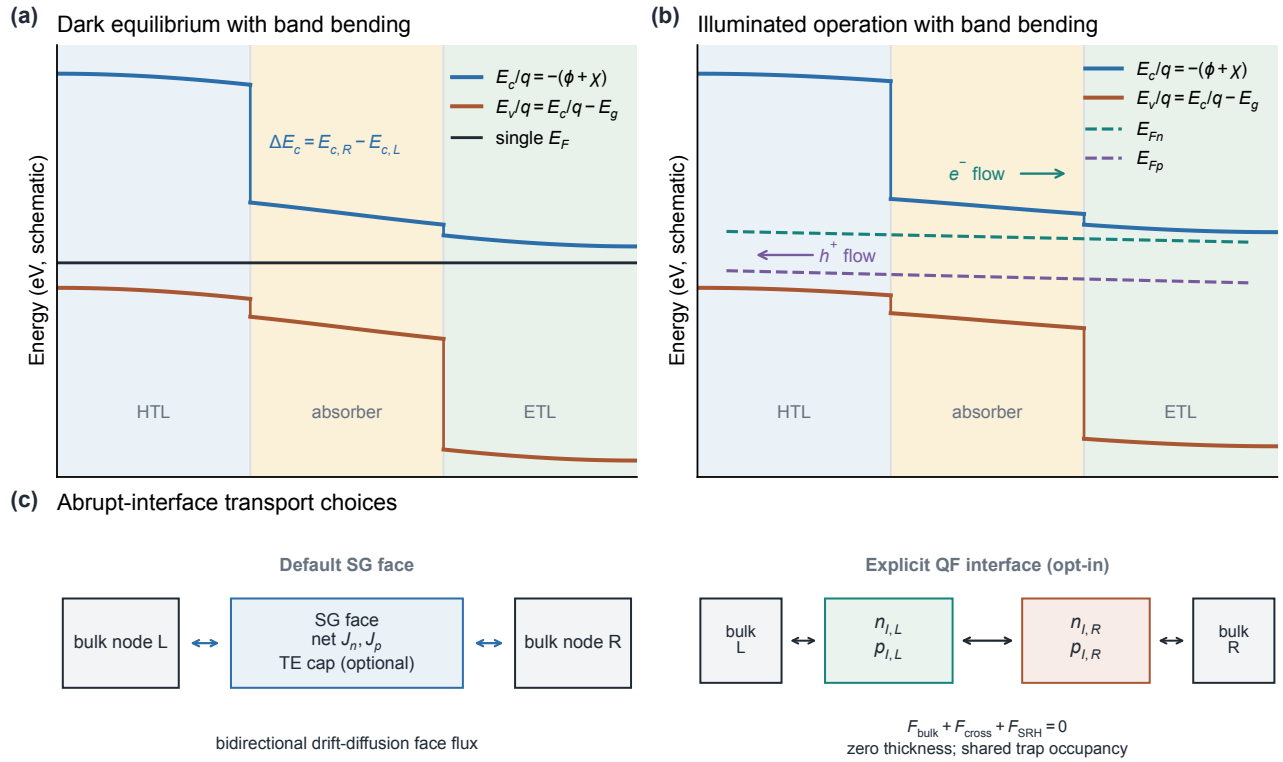


Figure 4.2: Band bending, quasi-Fermi levels, and abrupt-interface closures

The first two panels are schematic energy profiles rather than output from one particular device. They include a continuous electrostatic potential so that the bands bend within each layer, while the affinity and band-gap changes produce the abrupt offsets. At dark equilibrium there is one flat Fermi level. Under illumination the electron and hole quasi-Fermi levels split; the arrows show carrier motion toward the selective contacts, not the conventional-current sign. The default discretization uses one bidirectional Scharfetter-Gummel face whose magnitude may be limited by the optional thermionic cap. The opt-in QF boundary removes that face and balances bulk supply, reciprocal cross-interface transport, and shared interface SRH on a zero-thickness plane.

Chapter 5

Beginner Physics Primer

5.1 Electrons, Holes, And Bands

In a semiconductor, electrons can move in the conduction band and holes can move in the valence band. A solar cell works because light creates electron-hole pairs and the device structure makes it more likely that electrons leave through one contact while holes leave through the other.

SolarLab stores two energy parameters that are essential for heterojunctions:

- χ , the electron affinity;
- E_g , the band gap.

Together these determine approximate conduction-band and valence-band positions. At an interface, a change in χ or E_g creates band offsets. Band offsets are important because they can block one carrier type while allowing the other carrier type to pass.

5.2 Generation And Recombination

Optical generation $G(x)$ is the rate at which absorbed photons create electron-hole pairs per unit volume. Recombination $R(n, p)$ is the rate at which electrons and holes annihilate. A good solar cell has high generation and carrier collection, but low recombination.

SolarLab includes several recombination channels:

- Shockley-Read-Hall (SRH) trap-assisted recombination;
- radiative recombination;
- Auger recombination;
- interface recombination through surface recombination velocities;
- optional trap-density profiles near interfaces.

5.3 Mobile Ions And Hysteresis

Perovskite cells can contain mobile ionic defects. These defects move much more slowly than electronic carriers. During a voltage scan, the ion profile may not reach equilibrium at each voltage. This creates J-V hysteresis. SolarLab treats this as a physical transient, not as a post-processing correction.

In the state vector, the default mobile ion variable is P , the positive vacancy density. A second negative species can be configured through `D_ion_neg`, `P0_neg`, and `P_lim_neg`.

Chapter 6

Device Definition

6.1 Core Data Flow

The simulator data flow is:

YAML or inline device dictionary
-> MaterialParams + LayerSpec + DeviceStack
-> experiment driver
-> MaterialArrays cache
-> result dataclass

The main dataclasses are:

- MaterialParams: physical parameters for one material.
- LayerSpec: layer name, role, thickness, and material parameters.
- DeviceStack: full multilayer stack and global device settings.

6.2 Layer Roles

Layer roles used by the frontend and YAML schema are:

Role	Meaning
substrate	optical-only layer, excluded from electrical drift-diffusion
front_contact	front electrode or transparent conductor
ETL	electron transport layer
absorber	active photovoltaic absorber
HTL	hole transport layer
back_contact	rear electrode

Substrate layers must form a contiguous prefix of the stack. They participate in transfer-matrix optics but do not appear in the drift-diffusion grid.

6.3 Main Material Parameters

Field	Meaning	Unit
eps_r	relative permittivity	dimensionless
mu_n, mu_p	electron and hole mobilities	$m^2 V^{-1} s^{-1}$
v_th	layer thermal velocity used by interface kinetics	$m s^{-1}$
D_ion	positive ion diffusion coefficient	$m^2 s^{-1}$
P_lim	positive-ion finite-site density scale	m^{-3}
P0	initial positive ion density	m^{-3}
ni	intrinsic carrier density	m^{-3}
tau_n, tau_p	SRH lifetimes	s
n1, p1	SRH trap reference densities	m^{-3}
B_rad	radiative recombination coefficient	$m^3 s^{-1}$
C_n, C_p	Auger coefficients	$m^6 s^{-1}$
alpha	Beer-Lambert absorption coefficient	m^{-1}
N_A, N_D	acceptor and donor densities	m^{-3}
chi	electron affinity	eV
Eg	band gap	eV
Nc300, Nv300	effective band-edge densities of states at 300 K	m^{-3}
A_star_n, A_star_p	electron and hole Richardson constants	$A m^{-2} K^{-2}$
optical_material	key into data/nk/*.csv	string
incoherent	incoherent TMM layer flag	boolean

Additional optional parameters cover negative ions, temperature scaling, field-dependent mobility, trap profiles, and optical fallbacks.

6.4 Parameter Dictionary

Input specification is a dominant source of simulation error. SolarLab cannot infer whether a parameter comes from measurement, literature, fitting, or exploratory screening, so each field should be recorded with its source and value.

6.4.1 Electrical And Transport Fields

Field	Unit	Physical role	Beginner guidance
eps_r	1	relative permittivity in Poisson's equation	Larger values screen charge and reduce field gradients. Use layer-specific literature values rather than a single generic perovskite value.
mu_n	$m^2 V^{-1} s^{-1}$	low-field electron mobility	Controls electron extraction and transport resistance. Remember that $1 \text{ cm}^2 V^{-1} s^{-1} = 10^{-4} m^2 V^{-1} s^{-1}$.
mu_p	$m^2 V^{-1} s^{-1}$	low-field hole mobility	Same unit conversion convention as mu_n. Low mobility typically affects FF before producing a large change in V_{oc} .

Field	Unit	Physical role	Beginner guidance
v_th	$m s^{-1}$	carrier thermal velocity supplied to SCAPS-style interface kinetics	The default is $10^5 m s^{-1}$. The SCAPS loader imports the declared per-layer value; do not confuse it with drift velocity or mobility.
N_A	m^{-3}	ionized acceptor density	Represents p-type doping. Values reported in cm^{-3} must be multiplied by 10^6 .
N_D	m^{-3}	ionized donor density	Represents n-type doping. Use compensated doping only when it is part of the intended physical model.
ni	m^{-3}	intrinsic carrier density	Appears in mass action and SRH terms. It should be consistent with the effective band gap and temperature when doing parameter studies.
chi	eV	electron affinity	Sets approximate conduction-band alignment. Inconsistent χ values can introduce artificial transport barriers.
Eg	eV	band gap	Sets band offsets and thermodynamic interpretation. Beer-Lambert optics do not automatically shift spectral absorption when Eg changes.
A_star_n, A_star_p	$A m^{-2} K^{-2}$	electron and hole Richardson supplies for interface-limited transport	The physical interface uses one reciprocal prefactor per carrier, selected conservatively from the adjacent layers. Keep these values consistent with the effective DOS or effective-mass assumption.

6.4.2 Ion And Hysteresis Fields

Field	Unit	Physical role	Beginner guidance
D_ion	$m^2 s^{-1}$	positive mobile-ion diffusion coefficient	Set to zero when mobile ions are excluded from the model. Nonzero values can produce scan-rate dependence.
P0	m^{-3}	initial positive ion density	Represents the equilibrium mobile-vacancy population before redistribution.
P_lim	m^{-3}	positive-ion finite-site density entering the crowding chemical potential	Use a physically motivated mobile-site density. It is not a hard transient-state projection.
D_ion_neg	$m^2 s^{-1}$	negative mobile-ion diffusion coefficient	Enables a second mobile species. Leave zero for the default single-ion model.
P0_neg	m^{-3}	initial negative ion density	Use only when the negative species is part of the physical hypothesis.
P_lim_neg	m^{-3}	negative-ion finite-site density scale	Same interpretation as P_lim; a shared-site model normally uses a common reservoir scale.
E_a_ion	eV	Arrhenius activation energy for ion diffusion	Used for temperature-dependent ion mobility. Fitted or literature-derived values should be reported with provenance.

6.4.3 Recombination And Trap Fields

Field	Unit	Physical role	Beginner guidance
tau_n	s	electron SRH lifetime	Shorter lifetime increases trap-assisted recombination. It is often an effective fitted parameter.
tau_p	s	hole SRH lifetime	Interpret with tau_n; asymmetry can model carrier-selective trap response.
n1	m^{-3}	SRH electron reference density	Encodes the effective trap-energy position and should be changed consistently with the trap model.
p1	m^{-3}	SRH hole reference density	Companion to n1.
B_rad	$m^3 s^{-1}$	radiative recombination coefficient	Central to radiative-limit and photon-recycling studies.
C_n	$m^6 s^{-1}$	electron Auger coefficient	Most relevant when high carrier densities are expected.
C_p	$m^6 s^{-1}$	hole Auger coefficient	Same caution as C_n.
trap_N_t_interface	m^{-3}	interface-near trap density	Activates spatial trap profiles when supplied with decay information.
trap_N_t_bulk	m^{-3}	bulk trap density	The asymptotic trap density away from the interface.
trap_decay_length	m	decay length or Gaussian width	Must be physically resolvable by the grid.
trap_profile_shape	string	exponential or gaussian trap profile	Select the profile shape that corresponds to the assumed defect distribution.

6.4.4 Optical Fields

Field	Unit	Physical role	Beginner guidance
alpha	m^{-1}	scalar Beer-Lambert absorption coefficient	Suitable for simplified optical studies. It is not wavelength-resolved.
optical_material	string	key into tabulated n, k data	Required for TMM, EQE, and EL. The key must exist in perovskite_sim/data/nk.
n_optical	1	constant refractive-index fallback	Useful for approximate optical calculations but not a substitute for measured n, k .
incoherent	boolean	thick-layer incoherent TMM treatment	Intended for the first thick substrate. The current TMM path does not support arbitrary incoherent layers mid-stack.

6.4.5 Temperature And Field-Dependent Mobility Fields

Field	Unit	Physical role	Beginner guidance
Nc300	m^{-3}	conduction-band density of states at 300 K	Optional temperature-scaling input. Use only when the temperature model is calibrated.
Nv300	m^{-3}	valence-band density of states at 300 K	Companion to Nc300.

Field	Unit	Physical role	Beginner guidance
mu_T_gamma	1	mobility power-law temperature exponent	Default keeps a common scattering-style trend. Document any changed value.
B_rad_T_gamma	1	radiative coefficient temperature exponent	Default zero preserves legacy behavior.
varshni_alpha	eV K ⁻¹	Varshni band-gap coefficient	Zero disables band-gap temperature shift.
varshni_beta	K	Varshni temperature parameter	Used with varshni_alpha.
v_sat_n	$m s^{-1}$	electron velocity-saturation limit	Zero disables Caughey-Thomas saturation for electrons.
v_sat_p	$m s^{-1}$	hole velocity-saturation limit	Zero disables Caughey-Thomas saturation for holes.
ct_beta_n	1	electron Caughey-Thomas exponent	Controls sharpness of velocity saturation.
ct_beta_p	1	hole Caughey-Thomas exponent	Same role for holes.
pf_gamma_n	$(V/m)^{-1/2}$	electron Poole-Frenkel mobility coefficient	Zero disables field-enhanced hopping for electrons.
pf_gamma_p	$(V/m)^{-1/2}$	hole Poole-Frenkel mobility coefficient	Zero disables field-enhanced hopping for holes.

6.4.6 Device-Level Fields

Field	Unit	Physical role	Beginner guidance
built_in_potential_mode	string	source of the signed Poisson contact potential	Use semiconductor_work_function or metal_work_function for new physical configurations.
work_function_left_eV, work_function_right_eV	eV	left and right metal work functions below vacuum	Required together in metal_work_function mode.
V_bi_override	V	legacy manual magnitude	Valid only in legacy_manual; use it for a frozen benchmark, not as a fitted default.
V_bi	V	deprecated compatibility input	Retained so shipped and imported benchmark files keep their recorded boundary condition.
Phi	$m^{-2} s^{-1}$	incident photon flux for Beer-Lambert generation	For spectral experiments use TMM data rather than only changing Phi.
T	K	device temperature	Affects thermal voltage and enabled temperature-dependent hooks.
mode	string	legacy, fast, or full physics tier	The mode is a ceiling; hooks still need required parameters.
electrical_grid	object	optional per-layer interval weights and tanh clustering strengths	Both maps must cover every electrical layer exactly. The global N_grid budget is retained.
jv_solver_policy	string	production J-V variable-set requirement	general permits the established drivers; cancellation_safe_qf_required rejects an uncertified driver unless a diagnostic override is explicit.

Field	Unit	Physical role	Beginner guidance
interfaces	$m\ s^{-1}$ pairs	interface recombination velocities	One pair per internal electrical interface, ordered from the front contact toward the rear contact.
S_n_left	$m\ s^{-1}$	left electron contact velocity	None gives ohmic default. 0 is blocking. Large values approach ohmic extraction.
S_p_left	$m\ s^{-1}$	left hole contact velocity	Same convention as S_n_left.
S_n_right	$m\ s^{-1}$	right electron contact velocity	Same convention as S_n_left.
S_p_right	$m\ s^{-1}$	right hole contact velocity	Same convention as S_n_left.
microstructure	object	2D grain-boundary specification	Used by 2D drivers; ignored by standard 1D paths.

6.4.7 Advanced Model Options

These public DeviceStack fields change the equations or determine which solver paths are available. The defaults below apply to a new device. The legacy tier changes a few of them when an older benchmark must be reproduced without altering its original setup.

Field	Default	Effect
dos_band_potentials	true	folds per-layer effective DOS into the Boltzmann transport potentials; legacy mode disables it
te_physical_norm	false	replaces the empirical density-weighted TE cap with a dimensionally normalized emission-velocity form; it remains opt-in while convergence is being extended for strongly binding interfaces
ion_steric_diffusion_only	true	selects the modified-PNP lattice-gas flux of Eq. 7.14; false selects the legacy whole-flux regularization
ion_steric_shared_site	true	couples positive and negative mobile species through one finite-site reservoir when dual-ion mode is active
flat_band_contacts	false	activates the historical finite-rate Robin contact path; with an explicit potential mode it does not select the Poisson voltage
flat_band_metal_contacts, contact_phi_B_eV	false, 0	optional majority-reservoir floor for weakly doped contacts; the barrier input is separate from the two explicit metal work functions
interface_plane_projection	false	uses the interface-sampling projection from older SCAPS comparisons; the physical QF interface uses a separate boundary formulation
interface_two_sided	false	older two-sided cross-carrier interface capture approximation
interface_shared_occupancy	false	older shared-occupancy interface-state approximation
interface_plane_closure	false	older supply-limited QSS plane-recombination support while retaining established SG transport
interface_plane_generation	false	allows signed generation on the older plane closure when its detailed-balance reference is physical
het_recomb_despike	0	SCAPS-emulation blend for an interface-node bulk-recombination spike; zero keeps the unmodified physical discretization
band_grading	false	activates the declared per-layer Eg_back/chi_back grading law and associated mesh refinement
interface_tunneling, tunnel_mass_eff	false, 0.2	enables the static Padovani-Stratton enhancement at TE-capped faces and sets its effective mass

The QF driver owns `interface_boundary`, `interface_transport_model`, and the QF coordinate choice; they are not `DeviceStack` fields. These settings activate the boundary in Eq.~7.25 only for `quasi_fermi`. An unsupported combination raises an error. `perovskite_sim/models/config_loader.py` defines the accepted YAML keys; the table lists public physical options and omits private cache fields.

6.5 Global Device Parameters

The device-level YAML maps to the following `DeviceStack` fields. The legacy manual voltage is the only field whose public YAML name differs from its runtime storage name:

Field	Meaning
<code>built_in_potential_mode</code>	<code>semiconductor_work_function</code> , <code>metal_work_function</code> , or <code>legacy_manual</code>
<code>work_function_left_eV</code> , <code>work_function_right_eV</code>	explicit metal work functions for the two contacts
<code>V_bi_override</code> (YAML) $\rightarrow$ <code>V_bi</code> (runtime)	manual magnitude used only by the explicit legacy mode
<code>V_bi</code> (YAML)	deprecated alias retained for unmodified compatibility presets
<code>Phi</code>	incident photon flux for Beer-Lambert generation
<code>interfaces</code>	interface recombination velocities (v_n, v_p)
<code>T</code>	device temperature in K
<code>mode</code>	legacy, fast, or full
<code>grid_interval_weights</code> , <code>grid_alphas</code>	optional electrical-grid allocation and per-layer clustering controls
<code>jv_solver_policy</code>	<code>general</code> or <code>cancellation_safe_qf_required</code>
<code>S_n_left</code> , <code>S_p_left</code> , <code>S_n_right</code> , <code>S_p_right</code>	selective-contact velocities
<code>microstructure</code>	optional 2D grain-boundary block

6.6 Built-In Potential

In a new physical device file, SolarLab calculates the built-in potential from the equilibrium contact-potential difference,

$$V_{\text{bi}} = \phi(L) - \phi(0) = W_{\text{left}} - W_{\text{right}},$$

where each W is measured downward from vacuum and entered as a positive energy. `built_in_potential_mode` selects how W_{left} and W_{right} are supplied:

- `semiconductor_work_function` derives each contact work function from the outer electrical layer. It requires positive χ , E_g , $N_{C,300}$ and $N_{V,300}$ on both contact layers, together with finite non-negative doping. For Maxwell-Boltzmann statistics,

$$W_n = \chi + V_T \ln\left(\frac{N_C}{n}\right), \quad W_p = \chi + E_g - V_T \ln\left(\frac{N_V}{p}\right),$$

with n or p obtained from charge neutrality and mass action. The active temperature law supplies $N_C(T)$, $N_V(T)$ and the optional Varshni shift; contact-face doping and band grading are included. Loading stops if the DOS or band data needed for this calculation are missing.

- `metal_work_function` uses the measured or specified electrode values `work_function_left_eV` and `work_function_right_eV` directly. Use this mode when the electrodes are specified as metals.
- `legacy_manual` accepts a non-negative `V_bi_override` magnitude. The field direction still follows the endpoint junction orientation. Use it only for a frozen benchmark or imported protocol in which the manual voltage is part of the reference definition. New physical devices should use one of the two work-function modes.

SolarLab keeps the sign of the result. It is positive for the usual p-contact-left orientation and negative for an n-contact-left stack such as an n-window/p-absorber CIGS cell or an n^+/p silicon junction. Do not apply `abs()` at the Poisson boundary. The applied-voltage polarity is handled separately, so positive forward bias reduces the field magnitude in either orientation:

$$\phi(0) = \phi_{\text{left}}, \quad \phi(L) = \phi_{\text{left}} + V_{\text{bi}} - \text{sgn}(V_{\text{bi}})V_{\text{app}}.$$

With an explicit potential mode, `flat_band_contacts` affects only finite-rate Robin carrier exchange at the two outer contacts; the four `S_*` fields refine those rates. It leaves the Poisson voltage unchanged. The selected contact densities and potential must still be checked for dark-equilibrium consistency.

With doping-derived ohmic pins, a common dark quasi-Fermi level requires

$$V_{\text{bi}} = W_{s,\text{left}} - W_{s,\text{right}},$$

where the semiconductor work functions W_s use the same contact-face bands, DOS, doping and temperature as the pinned densities. `semiconductor_work_function` meets this condition by construction. In `metal_work_function` mode, the reservoir or Schottky barrier must also agree with the stated metal work function and adjacent band edge. Ordinary doping pins are suitable only when they describe the same effective reservoir; otherwise $V_{\text{app}} = 0$ can retain a contact quasi-Fermi mismatch even though the Poisson voltage is well defined. The optional `flat_band_metal_contacts` floor uses the separate `contact_phi_B_eV` input; it is not inferred from the two metal work functions.

Potential source	Carrier boundary	Interpretation
<code>semiconductor_work_function</code>	doping-derived ohmic pins	recommended ideal semiconductor-reservoir pair; dark-equilibrium compatible under the stated Maxwell-Boltzmann assumptions
<code>metal_work_function</code>	work-function-consistent Robin/Schottky reservoir	physical metal-contact pair when the barrier and exchange velocities have independent provenance

Potential source	Carrier boundary	Interpretation
metal_work_function	unchanged doping-derived pins	valid only if the two reservoirs are demonstrably equivalent; otherwise diagnostic or calibrated rather than an automatic equilibrium
legacy_manual	historical benchmark contacts	use only to reproduce the recorded benchmark; it does not define a new physical contact model

For a new contact model, verify zero dark current, spatially constant equilibrium quasi-Fermi levels and mesh convergence before using illuminated figures of merit.

Older files keep their original loading behavior. A device block containing `V_bi` but no `built_in_potential_mode` follows the historical compatibility rule, including the old `flat_band_contacts` implication. The shipped IonMonger and Driftdiffusion benchmarks therefore retain their recorded manual values. Saving the file without changing the selector preserves its old shape; selecting an explicit physical mode removes the manual field. For example, the IonMonger reference records 1.10 V, while its incomplete legacy band construction gives about 0.86 V. Substituting 0.86 V would define a different benchmark.

6.7 YAML Example

```
device:
  built_in_potential_mode: metal_work_function
  work_function_left_eV: 5.20
  work_function_right_eV: 4.10
  Phi: 2.5e21
  T: 300.0
  mode: full
  contacts:
    left:
      S_p: 1.0e3
      S_n: 1.0e-3
    right:
      S_n: 1.0e3
      S_p: 1.0e-3

electrical_grid:
  interval_weights:      # one positive weight per electrical layer
    HTL: 1.0
  alphas:                # per-layer tanh clustering strength
    HTL: 3.0

layers:
  - name: HTL
    role: HTL
```

```

thickness: 2.0e-7
eps_r: 3.0
mu_n: 1.0e-8
mu_p: 1.0e-6
ni: 1.0e9
N_A: 1.0e24
N_D: 0.0
D_ion: 0.0
P_lim: 1.0e27
P0: 0.0
tau_n: 1.0e-6
tau_p: 1.0e-6
n1: 1.0e9
p1: 1.0e9
B_rad: 0.0
C_n: 0.0
C_p: 0.0
alpha: 0.0
chi: 2.2
Eg: 3.0

```

For semiconductor-derived contacts, set `built_in_potential_mode` to `semiconductor_work_function` and provide `Nc300` and `Nv300` on both outer electrical layers. A frozen manual benchmark should state its exception rather than look like a physical default:

```

device:
  built_in_potential_mode: legacy_manual
  V_bi_override: 1.10

```

The old key `V_bi` remains readable only for unmodified compatibility files.

YAML 1.1 can parse bare scientific notation such as `1e-9` as a string. SolarLab coerces numeric-looking strings to floats, but writing values as `1.0e-9` is clearer and safer.

Chapter 7

Governing Equations

This chapter states the governing equations in dimensional SI form. The model belongs to the class of coupled Poisson/drift-diffusion (van Roosbroeck) systems, extended by Nernst-Planck transport for mobile ionic defects, heterojunction band alignment, and wavelength-resolved optical generation. The implementation evaluates discretized node and face arrays (Chapter 10), but the continuous form stated here clarifies the model assumptions and the physical meaning of each state variable.

Throughout, q is the elementary charge, k_B Boltzmann's constant, T the lattice temperature, and $V_T = k_B T / q$ the thermal voltage. Bulk carrier statistics are Maxwell-Boltzmann (non-degenerate). The opt-in physical interface boundary alone uses bounded Fermi occupations at the two band edges to keep its zero-thickness plane states below their available densities of states; this does not turn the bulk model into a Fermi-Dirac transport model.

7.1 Electrostatics: The Poisson Equation

The electrostatic potential $\phi(x, t)$ is obtained from the quasistatic Poisson equation

$$\frac{\partial}{\partial x} \left(\varepsilon_0 \varepsilon_r \frac{\partial \phi}{\partial x} \right) = -\rho, \quad (7.1)$$

with the space-charge density assembled from every charged species in the model:

$$\rho = q [p - n + (P - P_0) - (P^- - P_0^-) + N_D - N_A]. \quad (7.2)$$

The mobile-ion contributions are measured relative to their neutral background densities P_0 and P_0^- : an undisturbed vacancy population together with its immobile counter-charge is electrically neutral, so only the *redistribution* of ions contributes net space charge. The negative-species term is present only when a second mobile species is configured (dual-ion mode).

The quasistatic approximation asserts that the potential responds instantaneously to the charge configuration — Eq. 7.1 is Gauss's law $\nabla \cdot (\varepsilon_0 \varepsilon_r \mathbf{E}) = \rho$ with $E = -\partial \phi / \partial x$, solved at every instant of the transient. Retardation, magnetic effects, and electromagnetic wave propagation are excluded from the electrical solve (the optical field is treated separately by the transfer-matrix machinery of the *Optical Generation* section).

The discretization is finite-volume with a harmonic-mean face permittivity:

$$\varepsilon_{i+1/2} = \frac{2\varepsilon_i\varepsilon_{i+1}}{\varepsilon_i + \varepsilon_{i+1}}. \quad (7.3)$$

When the material interface falls at the midpoint between the two nodes, the harmonic mean is the exact series composition of the two dielectric half-cells: it is the face permittivity for which the discrete flux reproduces a continuous normal displacement $D = \varepsilon_0\varepsilon_r E$ across the discontinuity, with the field jumping between the layers as it physically must. An arithmetic (nodal) mean is the *parallel* composition instead, which is the correct average for a face-tangential field, so using it here misallocates the potential drop between the two layers. Neither choice literally imposes a boundary condition on E ; the difference is which equivalent series-versus-parallel dielectric response the face flux represents.

For a general non-symmetric geometry, where the interface does not sit at the midpoint, the correct series composition is the *distance-weighted* harmonic mean

$$\varepsilon_f = \frac{\Delta x_L + \Delta x_R}{\Delta x_L/\varepsilon_L + \Delta x_R/\varepsilon_R},$$

which reduces to Eq. 7.3 for equal half-widths. SolarLab places every material interface *on* a grid node and assigns that node the permittivity of the layer to its right, so each face lies entirely within one material and Eq. 7.3 smears the discontinuity over one half-cell rather than weighting two dissimilar half-cells. This is the usual node-centred finite-volume treatment and its error is first order in the local cell width; it is *not* a distance-weighting bug, but it does mean the interface is resolved to within half a cell, so interface-sensitive quantities should be checked for mesh convergence.

Assumptions and limits:

- electrostatics is quasistatic;
- magnetic effects and wave propagation in the electrical solve are ignored;
- layer interfaces are abrupt on the scale of the grid;
- ρ is built from the configured mobile species, carriers, and dopants, with dopants assumed fully ionized and immobile.

7.2 Carrier Transport: Continuity And Drift-Diffusion

The electron and hole densities obey the continuity equations

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G - R, \quad (7.4)$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G - R, \quad (7.5)$$

closed by the drift-diffusion constitutive relations

$$J_n = q\mu_n nE + qD_n \frac{\partial n}{\partial x}, \quad J_p = q\mu_p pE - qD_p \frac{\partial p}{\partial x}, \quad (7.6)$$

with the Einstein relation $D = \mu V_T$ linking diffusivity and mobility. The Einstein relation expresses local thermal equilibrium between the carrier gas and the lattice; it is exact for non-degenerate (Boltzmann) statistics.

An equivalent and physically transparent form writes each current as a quasi-Fermi-level gradient,

$$J_n = \mu_n n \frac{\partial E_{Fn}}{\partial x}, \quad J_p = \mu_p p \frac{\partial E_{Fp}}{\partial x}, \quad (7.7)$$

where, under Boltzmann statistics, $n = N_C \exp[-(E_C - E_{Fn})/k_B T]$ and $p = N_V \exp[-(E_{Fp} - E_V)/k_B T]$. Equations 7.6 and 7.7 are the same model; the quasi-Fermi form makes explicit that a flat quasi-Fermi level implies zero current *for that carrier* — the thermodynamic-equilibrium condition. At open circuit only the net terminal current is constrained to zero: internal electron and hole currents remain individually non-zero and convert into each other through generation and recombination, so the quasi-Fermi levels need not be flat everywhere in an operating device.

7.2.1 Heterojunction Band Alignment

Band edges are referenced to the local vacuum level through the electron affinity χ and the band gap E_g :

$$E_C/q = -(\phi + \chi), \quad E_V/q = -(\phi + \chi + E_g),$$

with χ and E_g expressed in volts (numerically equal to their eV values). In heterojunction stacks SolarLab therefore drives the carrier fluxes with *band-effective* potentials,

$$\phi_n = \phi + \chi, \quad \phi_p = \phi + \chi + E_g, \quad (7.8)$$

so that electrons respond to gradients of the conduction band edge and holes to gradients of the valence band edge. An abrupt change in χ or E_g between layers then produces Anderson-rule band offsets (ΔE_C from the affinity step, ΔE_V from the combined affinity-plus-gap step) inside the same numerical flux formulation, with no special-case interface treatment in the discretization.

7.2.2 Effective-Density-of-States Correction

For a heterostructure with unequal effective densities of states, the band-effective potentials of Eq. 7.8 are incomplete. Under Boltzmann statistics the carrier density couples to the quasi-Fermi level through the local N_C or N_V , so the exact drift potentials carry an additional logarithmic term:

$$\phi_n = \phi + \chi + V_T \ln \frac{N_C}{N_{C,\text{ref}}}, \quad \phi_p = \phi + \chi + E_g - V_T \ln \frac{N_V}{N_{V,\text{ref}}}. \quad (7.9)$$

Omitting these terms imposes a spurious quasi-Fermi-level discontinuity of $k_B T \ln(N_{C,1}/N_{C,2})$ (and the N_V analogue) at every interface with a density-of-states contrast — on a representative HTL/perovskite/ETL stack the accumulated error exceeds 100 mV of open-circuit voltage. SolarLab therefore folds the corrections of Eq. 7.9 into the cached per-layer χ/E_g arrays at build time, using the absorber as the reference layer (only cross-junction *differences* matter). The fold applies to

transport and thermionic emission only; purely statistical quantities (n_i , the SRH references n_1/p_1 , and the contact equilibrium densities) are left unchanged. The correction is active by default whenever per-layer Nc300/Nv300 data are supplied, and is disabled in the legacy tier, which reproduces the IonMonger convention without DOS-folded transport.

Assumptions and limits:

- carriers are described by drift-diffusion transport, not ballistic transport;
- carrier statistics are non-degenerate (Maxwell-Boltzmann); degenerately doped layers would require Fermi-Dirac corrections that are out of scope;
- the model is most natural when local quasi-equilibrium is a reasonable approximation;
- strongly quantum-confined structures, tunneling-dominated contacts, and hot carrier effects require additional modeling beyond this implementation.

7.3 Heterointerface Thermionic Emission And Tunnelling

The drift-diffusion picture assumes that the band edges vary smoothly on the scale of a grid cell. At an abrupt heterojunction the band offset is instead resolved within a *single* cell. With the band offsets and density-of-states corrections carried in the exponentially fitted potentials, the Scharfetter-Gummel flux (Chapter 10) is then the ideal-interface limit: it assumes local equilibrium across the step and unlimited interface transmission, i.e. no additional kinetic resistance at the junction. That is an upper bound on the transport a real interface supports whenever interface scattering, a finite emission velocity, tunnelling, or interface-state supply actually limit it — it is not a systematic discretization error for every band step, and the size of the discrepancy is a property of the interface, not of the scheme. SolarLab therefore bounds the drift-diffusion face flux by an interface-limited thermionic-emission (TE) flux of Richardson-Dushman form at every face where the conduction- or valence-band offset exceeds 50 meV:

$$J_{\text{TE}} \propto A^* T^2 \left[n_L e^{-\max(\Delta E, 0)/V_T} - n_R e^{-\max(-\Delta E, 0)/V_T} \right], \quad (7.10)$$

where ΔE is the band-edge step across the face, n_L and n_R are the adjacent carrier densities, and A^* is a per-face Richardson constant (default: the free-electron value $1.2017 \times 10^6 \text{ A m}^{-2} \text{ K}^{-2}$, overridable per layer). Only the *uphill* term carries the Boltzmann barrier penalty; a downhill offset leaves the emission term unpenalized, so carriers flowing down a band step are not artificially blocked.

Two points are important when using the default form. The density-weighted form of Eq. 7.10 is an *empirical interface-limited bound*. It is not the dimensional Richardson-Dushman current: $A^* T^2$ is already a current density, so multiplying by an absolute carrier density leaves the bound's magnitude tied to the SI unit system. This makes $|J_{\text{TE}}|$ very large (order 10^{28} – 10^{35} on a perovskite stack), so the smaller-magnitude cap is rarely active; the observed heterojunction V_{oc} behaviour is driven by the band-offset potentials of Eqs. 7.8–7.9, not by the TE cap. A dimensionally correct *emission-velocity* form is available as an opt-in (te_physical_norm): dividing Eq. 7.10 by the band-edge density of states converts $A^* T^2$ into an emission velocity $v_R = A^* T^2 / (q N_C)$, giving $J = q v_R(\dots)$, which binds at real interface densities. The normalized cap gives $V_{\text{oc}} = 1.2013 \text{ V}$ on the mirrored regression configuration, the same value as the legacy cap after correcting the cap direction and barrier input. It remains opt-in because the coupled density solve still loses convergence in some strongly binding cases, especially with weak contacts, difficult cold starts or deep forward injection. The limitation is in the current density solver, not in the dimensional normalization. This option applies only to the bulk-node transport cap used by the transient and direct steady-state drivers; interface-plane transport uses its own flux law. Configurations without per-layer effective-DOS data are bit-identical with the flag on

or off. The second point concerns equilibrium. The two-leg bracket vanishes only when the adjacent layers share the same effective density of states or when the density-of-states-folded potentials of Eq. 7.9 are active. The *cap construction* still preserves system-level equilibrium: the drift-diffusion flux is exactly zero, so the smaller-magnitude selection below also returns zero for any value of J_{TE} .

The TE flux caps magnitude only. At each flagged face, the solver takes the smaller magnitude and keeps the direction of the drift-diffusion flux. The interface limit can reduce the predicted current, but cannot enhance or reverse it. The thermionic expression specifies how much current the interface can emit; the underlying drift-diffusion solution specifies its direction. Using the independent sign of the bracket in Eq. 7.10 could reverse that current. The steady-state Newton driver replaces the hard minimum with a smooth sigmoid blend of adjustable width to restore differentiability; the blend likewise interpolates magnitudes. See Chapter 10.

7.3.1 Thermionic-Field-Emission Tunnelling

Pure thermionic emission transports carriers *over* the barrier only. An optional intra-band tunnelling channel augments it with transport *through* the top of a band spike, as a **doping-derived static tunnelling enhancement** scaled by the Padovani-Stratton characteristic energy E_{00} . It is not an implementation of the Padovani-Stratton thermionic-field-emission current: that theory evaluates a current-voltage relation from an energy integral over a specific barrier shape, whereas the enhancement below is a single bias-independent factor derived from the doping. In particular it does not solve for the bias-dependent depletion width, the local field, or the barrier profile, and it is therefore not equation-level equivalent to the WKB tunnelling model in SCAPS (which computes a transmission probability with a numerical floor on the barrier height and enters the DC solution only). Cross-simulator comparisons should either disable tunnelling on both sides or benchmark the two channels separately against a common barrier and effective mass. The characteristic tunnelling energy is

$$E_{00} = \frac{\hbar}{2} \sqrt{\frac{N_{\text{if}}}{m^* \varepsilon_s}}, \quad E_0 = E_{00} \coth\left(\frac{E_{00}}{V_T}\right), \quad (7.11)$$

where E_{00} is expressed in eV (the prefactor $\hbar/2$, without the charge factor, yields volts directly; the implementation computes the joule value $(q\hbar/2)\sqrt{N_{\text{if}}/(m^*\varepsilon_s)}$ and divides by q , which is the same number), so the ratio E_{00}/V_T in the coth is dimensionally consistent. N_{if} is the net doping of the lighter-doped (depletion) side of the junction, m^* the effective tunnelling mass (default $0.2 m_e$), and ε_s the semiconductor permittivity. The tunnelling-transparent fraction of the barrier, $\delta_{\text{tun}} = |\Delta E| (1 - V_T/E_0)$, clamped to $[0, |\Delta E|]$, defines a static enhancement factor

$$\Gamma = \exp\left(\frac{\delta_{\text{tun}}}{V_T}\right) \geq 1, \quad (7.12)$$

which is folded into the per-face Richardson constants ($A_{\text{eff}}^* = \Gamma A^*$) at build time. Because the same Γ scales *both* legs of Eq. 7.10, equilibrium detailed balance is preserved to machine precision, and because the factor is static (doping-derived, no dependence on the evolving field) it cannot perturb the stability of the time integration. The channel is opt-in and captures the doping-direction trend of tunnelling-assisted interface transport; the bias-dependent flattening of a spike under forward bias would require a field-dependent $E_{00}(F)$ evaluated per step, which is outside the present scope.

Assumptions and limits:

- TE capping applies only at faces whose band offset exceeds 50 meV; smaller offsets are handled by the drift-diffusion flux alone;
- the tunnelling enhancement is static and doping-derived; an intrinsic-absorber interface ($N_{\text{if}} \rightarrow 0$) gives $E_{00} \rightarrow 0$ and hence $\Gamma \rightarrow 1$ (no enhancement);
- the legacy tier disables thermionic emission entirely (IonMonger has no TE), and with it the tunnelling channel by construction.

7.4 Ion Migration

Mobile ionic defects — in halide perovskites, predominantly halide vacancies acting as an effective positive species — are transported by a Nernst-Planck equation of the same drift-diffusion form as the electronic carriers. The positive vacancy density P satisfies the conservation law

$$\frac{\partial P}{\partial t} = -\frac{\partial F_P}{\partial x}, \quad (7.13)$$

The default is the finite-site modified Poisson-Nernst-Planck model. For mobile species s with number density P_s , charge number $z_s \in \{-1, +1\}$ and diffusivity D_s , its continuous flux is

$$F_s = -D_s \left[\frac{\partial P_s}{\partial x} + P_s \frac{\partial}{\partial x} \left(\frac{\mu_{\text{ex}}}{k_B T} \right) + z_s \frac{P_s}{V_T} \frac{\partial \phi}{\partial x} \right], \quad (7.14)$$

with the lattice-gas crowding potential

$$\frac{\mu_{\text{ex}}}{k_B T} = -\ln(1 - \theta). \quad (7.15)$$

For one positive species, $\theta = P/P_{\text{lim}}$ and Eq. 7.14 reduces to

$$F_P = -D_I \left[\frac{1}{1 - P/P_{\text{lim}}} \frac{\partial P}{\partial x} + \frac{P}{V_T} \frac{\partial \phi}{\partial x} \right]. \quad (7.16)$$

Thus finite-site crowding modifies the chemical-diffusion term without artificially multiplying the electrostatic drift. For two mobile species on a shared site reservoir, the default `ion_steric_shared_site` model uses $\theta = (P_+ + P_-)/P_{\text{lim}}$ in Eq. 7.15; both species then respond to the same vacancy fraction. Setting that flag false assigns distinct sublattices, with each species using its own occupancy and declared site-density scale. The charge sign enters only through z_s .

There is no generation or recombination term: the total number of mobile ions in the device is conserved. Consistently, the contacts are ion-blocking,

$$F_P(0) = F_P(L) = 0, \quad (7.17)$$

so that $\int_0^L P dx$ is an exact invariant of the dynamics — ions redistribute internally but never leave through the electrodes.

The implementation evaluates $-\ln(1 - \theta)$ with θ clipped to $[0, 0.999999]$ so a trial state cannot make the logarithm singular. This protects the flux evaluation; it does not project the transient state onto $P_s \leq P_{\text{lim}}$. P_{lim} is the finite-site density in the chemical potential, not a numerically enforced upper bound. Near-saturation results require an occupancy and time-step study.

7.4.1 Legacy Whole-Flux Steric Form

The legacy tier retains the pre-July-2026 compatibility law

$$F_P^{\text{legacy}} = -s(P)D_I \left(\frac{\partial P}{\partial x} + \frac{P}{V_T} \frac{\partial \phi}{\partial x} \right), \quad s(P) = \frac{1}{\max(1 - P/P_{\text{lim}}, 10^{-6})}. \quad (7.18)$$

Because the same positive factor multiplies diffusion and drift, this law rescales the kinetics but leaves the ideal Boltzmann zero-flux equilibrium unchanged; it is an empirical crowding regularization, not a finite-site occupation model. The shipped presets remain in the dilute regime ($P/P_{\text{lim}} \approx 0.011$ on the IonMonger benchmark), where the two forms differ by less than 0.1 mV in V_{oc} . At high occupancy the old law produced increasing FF and J_{sc} as the lattice filled, whereas the modified-PNP form remained stable across the registered occupancy sweep. That physical trend, while preserving the legacy benchmark path, is the reason the modified-PNP law is the current default.

Because ionic diffusivities ($D_I \sim 10^{-16} \text{ m}^2 \text{ s}^{-1}$) are many orders of magnitude below carrier diffusivities, ionic redistribution introduces a slow time scale that coexists with nanosecond electronic relaxation. The governing relaxation is the drift-assisted screening of the built-in field across the interfacial, Debye-scale charge layers — typically milliseconds to seconds at perovskite ion mobilities — rather than diffusion over the full device length, which would be slower still. This stiff two-time-scale structure is the physical origin of scan-rate-dependent J-V hysteresis, and it is what the implicit time integration of Chapter 10 is designed to handle.

Additional model features:

- a second, negatively charged mobile species is optional and obeys the same equations with the drift sign reversed;
- ionic diffusivity is Arrhenius-activated in temperature through the per-layer activation energy $E_{\text{a_ion}}$ when temperature scaling is enabled;
- non-ionic layers simply set $D_I = 0$; the ion equations still integrate but contribute zero flux.

Assumptions and limits:

- the default mobile species is a positive vacancy-like species;
- negative mobile species are optional and must be explicitly configured;
- contacts are ion-blocking in the implemented boundary condition;
- the 2D J-V solver treats ions as a frozen Poisson background.

7.5 Recombination

All bulk recombination channels share the mass-action driving term $np - n_i^2$, which vanishes at thermodynamic equilibrium and guarantees that no channel generates or destroys carriers in the dark equilibrium state.

7.5.1 Shockley-Read-Hall (Trap-Assisted) Recombination

$$R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}. \quad (7.19)$$

The reference densities n_1 and p_1 encode the trap energy level E_t through the Boltzmann relations

$$n_1 = N_C e^{-(E_C - E_t)/k_B T}, \quad p_1 = N_V e^{-(E_t - E_V)/k_B T}, \quad n_1 p_1 = n_i^2.$$

For approximately symmetric capture and at the injection levels typical of an operating cell, deep levels near midgap produce the strongest SRH recombination, because n_1 and p_1 are then smallest relative to n and p and the denominator is minimized. The statement is conditional in both respects: the rate-maximizing level depends on n , p , τ_n and τ_p , and shifts away from midgap for strongly asymmetric capture; and $n_1, p_1 \ll n, p$ is not guaranteed at midgap, since inside the depletion region or under low injection the carrier densities can fall to the order of n_i themselves. The asymmetric lifetimes $\tau_n \neq \tau_p$ model carrier-selective capture. In SolarLab the SRH parameters are *effective* quantities per layer unless trap-energy provenance is supplied.

7.5.2 Radiative Recombination And Photon Recycling

Band-to-band radiative recombination is bimolecular:

$$R_{\text{rad}} = B_{\text{rad}}(np - n_i^2). \quad (7.20)$$

In a high-refractive-index absorber most internally emitted photons are trapped by total internal reflection and reabsorbed, regenerating carriers — photon recycling. Following Yablonovitch's statistical ray optics, the single-pass escape probability of an internally emitted photon is

$$P_{\text{esc}} = \min\left(1, \frac{1}{4 n_r^2 \alpha(\lambda_g) d}\right), \quad (7.21)$$

evaluated at the band-edge wavelength $\lambda_g = hc/E_g$ from the absorber's tabulated n , k data, with d the absorber thickness. Only the escaping fraction is a true loss. SolarLab implements this at two levels of self-consistency:

1. **Net-coefficient form** (photon recycling on): the absorber's B_{rad} is rescaled by P_{esc} once at build time. Exact whenever the np product is spatially uniform across the absorber (the open-circuit regime where the V_{oc} boost is measured).
2. **Self-consistent reabsorption** (full tier): B_{rad} keeps its intrinsic value as a sink, and at every solver evaluation the total *net* emission $\int B_{\text{rad}}(np - n_i^2) dx$ over the absorber is redistributed as a uniform generation source $G_{\text{rad}} = (1 - P_{\text{esc}}) [\int B_{\text{rad}}(np - n_i^2) dx] / d$. The net-emission integrand makes the recycled source vanish at mass-action equilibrium together with the sink, so the channel preserves detailed balance. This closes the recycling loop and retains the *global* coupling when np is non-uniform (strong injection, near-contact gradients, tandem sub-cells); it reduces to form 1 in the uniform limit. It is, however, a *spatially averaged* approximation: the redistribution is uniform over the absorber, with no spatial or spectral reabsorption kernel (emission position, propagation direction, wavelength-dependent absorption depth, and parasitic reabsorption are not resolved).

On the canonical radiative-limit configuration the recycling boost falls within the 40–100 mV regression acceptance window adopted for that configuration (motivated by reported MAPbI₃ recycling effects, e.g. Pazos-Outón et al.). The recycling gain depends on the internal radiative efficiency, parasitic absorption, and out-coupling, so the window is a regression gate for the canonical preset rather than a general MAPbI₃ figure.

7.5.3 Auger Recombination

$$R_{\text{Auger}} = (C_n n + C_p p)(np - n_i^2), \quad (7.22)$$

the three-particle channel that dominates at high carrier density (heavily doped layers, concentrator conditions, band-offset accumulation spikes).

The total bulk rate is:

$$R = R_{\text{SRH}} + R_{\text{rad}} + R_{\text{Auger}}. \quad (7.23)$$

7.5.4 Interface Recombination

Each internal electrical interface carries an areal SRH channel parameterized by surface recombination velocities (v_n, v_p) :

$$R_s = \frac{\frac{n_s p_s - n_{i,\text{eff}}^2}{n_s + n_1} + \frac{p_s + p_1}{v_p}}{\frac{1}{v_p} + \frac{1}{v_n}} \quad [m^{-2}s^{-1}], \quad (7.24)$$

converted to a volumetric sink at the interface node by dividing by the local dual-cell width. Three details matter for heterojunctions:

- **Cross-carrier sampling.** At a defect-bearing heterointerface the electron density is sampled on the transport-layer side and the hole density on the absorber side of the junction — the carrier populations that actually accumulate under cliff-type band offsets, following the polycrystalline-heterojunction treatment of Pauwels and Vanhoute. This reproduces the characteristic cliff/spike asymmetry of interface-limited V_{oc} .
- **Detailed balance.** The reference $n_{i,\text{eff}}^2$ is the *product of the adjacent layers' equilibrium densities* ($n_{\text{eq},R} p_{\text{eq},L}$), not $n_{i,L} n_{i,R}$: with a heavily doped transport layer the equilibrium cross product exceeds the intrinsic product by many orders of magnitude, and a naive intrinsic reference would make the interface a spurious generator at dark equilibrium.
- **Non-negativity clamp (model-specific safeguard, defect-scoped).** At interfaces carrying a *declared defect* — where the cross-carrier approximation installs a bulk-asymptotic reference $n_{i,\text{eff}}^2$ that can exceed the true interface-plane product by orders of magnitude — the implementation clamps $R_s \geq 0$ to suppress the resulting spurious generation (of order 10 A/m² at illuminated short circuit). This clamp is *not* a general physical property: a real interface defect does generate carriers when $n_s p_s < n_{i,\text{eff}}^2$, the origin of depletion-region generation current and reverse dark current. Interfaces *without* a declared defect use the local mass-action reference and are left unclamped, so physical depletion generation is retained there.

The July 2026 audit found no resolvable physical generation removed on the mirrored comparison stack. The cross-carrier reference is the cross-side product of two *majority* equilibrium densities, which on the mirrored configuration is $1.0 \times 10^{44} \text{ m}^{-6}$ against a local mass-action n_i^2 of 1.98×10^{24} at the same node — twenty orders of magnitude larger. Lifting the clamp there yields -8.4 A m^{-2} at -0.5 V , but that figure is set by the inflated reference, not by interface physics. Evaluated instead against a true detailed-balance reference — the interface-plane closure's $n_{i,s}^2$, which equals the node's own n_i^2 on this stack — the signed rate is $-8 \times 10^{-21} \text{ A m}^{-2}$, with a ceiling near 4×10^{-12} even for a fully depleted plane. That numerical conclusion belongs to this stack and surface-velocity range; it is not a general reason to discard interface generation.

The older SCAPS-comparison options remain available and default off. They either evaluate Eq. 7.24 on Boltzmann-projected plane densities or solve a supply-limited plane closure with the reduced interface gap $E_{g,s} = \min(E_C) - \max(E_V)$. These options modify the recombination support while retaining the established transport formulation.

7.5.4.1 Exclusive Physical Interface Boundary

The quasi-Fermi driver provides a stricter alternative through `interface_boundary=true`. At every abrupt internal layer boundary, the ordinary Scharfetter-Gummel face is removed and replaced by a zero-thickness plane carrying four one-sided densities,

$$\mathbf{s}_{\text{if}} = (n_{R,s}, p_{R,s}, n_{L,s}, p_{L,s}).$$

These densities are not extra device-level dynamic unknowns. At each outer residual evaluation they are eliminated from the local steady balance

$$\mathbf{F}_{\text{bulk}} + \mathbf{F}_{\text{cross}} + \mathbf{F}_{\text{SRH}} = \mathbf{0}. \quad (7.25)$$

Only the reservoir-to-plane fluxes enter the two adjacent finite-volume carrier balances. Cross-interface exchange remains internal to the plane. The original SG face is zeroed, so the offset barrier is neither bypassed nor counted twice. The reservoir flux used by the outer balance is projected onto $-(\mathbf{F}_{\text{cross}} + \mathbf{F}_{\text{SRH}})$ to remove the round-off left by subtracting nearly equal thermal-velocity supplies; the unprojected mismatch is retained and must pass the final interface certificate. The projection therefore enforces conservation without hiding a failed local constitutive solve.

One shared trap occupancy couples both sides. With $i \in \{L, R\}$,

$$f = \frac{v_n \sum_i n_{i,s} + v_p \sum_i p_{1,i}}{v_n \sum_i (n_{i,s} + n_{1,i}) + v_p \sum_i (p_{i,s} + p_{1,i})}, \quad (7.26)$$

and the signed electron and hole capture rates on side i are

$$R_{n,i} = v_n [n_{i,s}(1 - f) - n_{1,i}f], \quad R_{p,i} = v_p [p_{i,s}f - p_{1,i}(1 - f)].$$

This construction gives $\sum_i R_{n,i} = \sum_i R_{p,i}$ and vanishes at thermal equilibrium without a non-negativity clamp. The local solve uses a DOS-bounded logit for the `fermi_richardson` model and logarithmic densities for the two Boltzmann models. An inexact inner value may guide the outer Newton iteration, but the final state is solved again and rejected if its local normalized residual exceeds 10^{-7} .

Three reciprocal cross-plane laws are selectable:

- `fermi_richardson` (default) evaluates the two one-way supplies as bounded band-edge Fermi occupations with one Richardson prefactor in both directions. It recovers Boltzmann thermionic emission in the dilute limit, enforces $\text{state/DOS} \leq 1$, and is the preferred law when a barrier drives a plane population toward its available DOS.
- `scaps_thermionic` implements the published SCAPS Boltzmann form at the common maximum band edge, with a common Richardson supply. It is a compatibility law; the CBO protocol accepts it only when the maximum state/DOS ratio is no greater than 0.1.
- `scaps_thermal_velocity` uses the smaller of the two adjacent layer v_{th} values with the DOS-correct equilibrium density step. It is also a Boltzmann compatibility law and carries the same 0.1 dilute-state restriction in the CBO certificate.

All three use the same prefactor in the forward and reverse directions and therefore give zero cross-plane current for a common quasi-Fermi level. The physical boundary requires positive per-layer N_C, N_V and physical band offsets; unsupported contacts, ions, field-dependent mobility, non-local photon recycling, and older interface-state combinations are rejected before Newton starts. It is opt-in and does not change the default transient, direct-steady-state, or quasi-Fermi homojunction paths.

A related optional correction (`het_recomb_despike`, a fraction $f \in [0, 1]$) addresses a discretization artifact: an abrupt valence-band offset produces a Boltzmann accumulation *spike* of majority carriers on the single interface node, and because $R_{\text{Auger}} \propto p^2 n$, the bulk Auger channel over-counts an interface loss that the areal channel of Eq. 7.24 already represents. The correction blends the junction-node densities toward the geometric mean of their neighbours *for the bulk-recombination evaluation only* — transport fluxes are untouched.

Assumptions and limits:

- SRH parameters are effective trap parameters unless trap-energy provenance is supplied;
- radiative and Auger coefficients should be layer-appropriate;
- interface recombination is localized numerically near the interface, so grid resolution matters when very large surface velocities are used;
- a grid-certified transition in an interface sweep is an internal numerical result until an independent physical curve validates its location and shape.

7.6 Optical Generation

SolarLab supports two optical models of increasing fidelity.

7.6.1 Beer-Lambert Absorption

The scalar model uses a single absorption coefficient per layer:

$$G(x) = \Phi \alpha(x) \exp \left[- \int_0^x \alpha(x') dx' \right]. \quad (7.27)$$

It captures exponential attenuation of a monochromatic-equivalent photon flux Φ but no interference, reflection, or spectral structure.

7.6.2 Transfer-Matrix Optics

For wavelength-resolved generation, SolarLab implements the coherent transfer-matrix method (TMM) for multilayer thin films at normal incidence, following Pettersson et al. and Burkhard et al. Each layer is described by its complex refractive index $\tilde{n}(\lambda) = n + ik$ from tabulated n, k data. For every wavelength, 2×2 *interface* matrices built from the Fresnel coefficients

$$r_{ab} = \frac{\tilde{n}_a - \tilde{n}_b}{\tilde{n}_a + \tilde{n}_b}, \quad t_{ab} = \frac{2\tilde{n}_a}{\tilde{n}_a + \tilde{n}_b},$$

alternate with *propagation* matrices carrying the complex phase $\delta_j = 2\pi\tilde{n}_j d_j / \lambda$ through each layer of thickness d_j . The assembled system matrix yields the stack reflectance and the forward/backward field amplitudes at every depth, from which the normalized internal intensity $|E(x, \lambda)|^2 / |E_0|^2$ follows. The local spectral absorption rate is

$$A(x, \lambda) = \frac{4\pi n(\lambda) k(\lambda)}{\lambda n_{\text{amb}}} \frac{|E(x, \lambda)|^2}{|E_0|^2} \quad [m^{-1}], \quad (7.28)$$

where the n/n_{amb} factor converts field energy density to the Poynting-vector (energy-flux) normalization; this is the normalization under which the computed reflectance, transmittance, and layer-resolved absorptance satisfy $R + T + A = 1$, which the regression suite checks explicitly. The generation profile is the spectral integral against the incident AM1.5G photon flux,

$$G(x) = \int A(x, \lambda) \Phi_{\text{AM1.5G}}(\lambda) d\lambda, \quad (7.29)$$

evaluated over 300–1000 nm by default, assuming unit internal quantum efficiency of photogeneration, and cached once before the transient solve.

A millimetre-scale glass substrate is far thicker than the coherence length of sunlight, so treating it coherently would produce unphysical interference fringes. The first layer of the stack may therefore be flagged *incoherent*: it contributes a phase-free power-form Fresnel reflection and Beer-Lambert bulk attenuation, and hands the transmitted intensity to the coherent sub-stack beneath it.

EQE and electroluminescence require this wavelength-resolved machinery (monochromatic TMM generation plus a short-circuit drift-diffusion solve per wavelength); Beer-Lambert-only stacks do not contain sufficient spectral information for those experiments.

Assumptions and limits:

- Beer-Lambert optics are fast and useful for trends, but cannot represent interference, reflection, or wavelength-dependent collection;
- the TMM assumes planar layers, normal incidence, and coherent propagation (except the optional incoherent first substrate);
- TMM quality is limited by the provenance and wavelength coverage of the supplied optical constants;
- synthetic or placeholder optical constants should be treated as workflow demonstrations, not material-specific evidence.

7.7 Field-Dependent Mobility

Two empirical high-field mobility models can modify the low-field mobilities, composed multiplicatively and evaluated per face from the Poisson-solved electric field at every solver evaluation. The Caughey-Thomas form imposes velocity saturation,

$$\mu_{\text{CT}}(E) = \mu_0 \left[1 + \left(\frac{\mu_0 |E|}{v_{\text{sat}}} \right)^\beta \right]^{-1/\beta}, \quad (7.30)$$

with $\beta = 2$ (the Canali form, typical for electrons) and $\beta = 1$ (the Thornber form, typical for holes) as conventional exponents. The Poole-Frenkel form describes field-assisted barrier lowering in disordered or organic transport layers,

$$\mu_{\text{PF}}(E) = \mu_0 \exp(\gamma_{\text{PF}} \sqrt{|E|}), \quad (7.31)$$

with $\gamma_{\text{PF}} \sim 3 \times 10^{-4} (\text{V}/\text{m})^{-1/2}$ representative of spiro-OMeTAD. Both hooks are opt-in per layer ($v_{\text{sat}} = 0$ and $\gamma_{\text{PF}} = 0$ disable them) and, because they must be re-evaluated from the instantaneous field, they are active only in the `full` tier.

7.8 Temperature Dependence

When temperature scaling is enabled, the device temperature T enters the model self-consistently. The thermal voltage $V_T = k_B T / q$ rescales every Einstein relation; carrier mobilities follow the standard scattering-style power law $\mu(T) = \mu_{300} (T/300 \text{ K})^{\gamma_\mu}$; the radiative coefficient follows $B_{\text{rad}}(T) = B_{300} (T/300 \text{ K})^{\gamma_B}$; and ionic diffusivity is Arrhenius-activated through the per-layer activation energy `E_a_ion`.

The band gap follows a Varshni shift *referenced to 300 K*, so that the configured E_g retains its meaning as the room-temperature gap:

$$E_g(T) = E_g^{300} - \alpha \left[\frac{T^2}{T + \beta} - \frac{T_0^2}{T_0 + \beta} \right], \quad T_0 = 300 \text{ K}. \quad (7.32)$$

Conventional semiconductors (silicon: $\alpha \approx 4.73 \times 10^{-4} \text{ eV/K}$, $\beta \approx 636 \text{ K}$) narrow on heating; halide perovskites show the opposite sign and widen with temperature, reproduced by a negative α .

The band-edge effective densities of states follow the standard parabolic-band law,

$$N_C(T) = N_C^{300} \left(\frac{T}{300 \text{ K}} \right)^{3/2}, \quad N_V(T) = N_V^{300} \left(\frac{T}{300 \text{ K}} \right)^{3/2}, \quad (7.33)$$

and the intrinsic density is recomputed from the shifted gap and the scaled densities of states, $n_i(T) = \sqrt{N_C(T) N_V(T)} e^{-E_g(T)/2k_B T}$. Because the SRH reference densities n_1, p_1 are configured at 300 K, they are rescaled by $n_i(T)/n_i(300)$ so that the mass-action invariant $n_1 p_1 = n_i^2(T)$ is preserved at every temperature; this holds the trap level fixed relative to midgap, the same convention used for

graded layers. Without that rescaling the SRH reference densities would stay pinned at their 300 K values while $n_i(T)$ moved, which biases the dark and depletion-regime rates that a $V_{oc}(T)$ sweep reads.

Two quantities remain fixed at their 300 K values. Both occur on interface paths that are off by default: the thermal velocity of the interface-plane state and closure formulations (fixed at the SCAPS value 10^7 cm s^{-1} rather than $v_{th} \propto T^{1/2}$), and the density-of-states prefactor of the per-side interface-defect trap levels. Cross-simulator $V_{oc}(T)$ comparisons that enable those options should hold $T = 300 \text{ K}$ or supply the temperature dependence explicitly.

All temperature hooks are gated by the temperature-scaling flag; the legacy tier pins the model at 300 K regardless of the configured T .

7.9 Spatial Trap Profiles

Grain-boundary and interface-adjacent defects concentrate near the transport layers. A per-layer trap-density profile with exponentially decaying edge enhancement,

$$N_t(x) = N_t^{\text{bulk}} + (N_t^{\text{if}} - N_t^{\text{bulk}}) (e^{-x/L_d} + e^{-(d-x)/L_d}), \quad (7.34)$$

(or a Gaussian variant with faster tail decay for defect slabs of well-defined width) is mapped onto the local SRH lifetimes through the inverse-proportionality of the SRH lifetime to trap density,

$$\tau(x) = \tau_{\text{bulk}} \frac{N_t^{\text{bulk}}}{\max(N_t(x), N_t^{\text{bulk}})},$$

where the denominator floor caps the lifetime at τ_{bulk} : an edge-*passivation* profile ($N_t^{\text{if}} < N_t^{\text{bulk}}$) is clamped to the bulk lifetime rather than extrapolated to lifetimes above τ_{bulk} , so the profile can only ever add recombination. The two edge kernels are summed rather than maximized so that overlapping tails blend smoothly, and the sum is divided by its analytic supremum so that the shape function is bounded by unity and N_t saturates at N_t^{if} . Without that normalization the raw sum reaches $N_t^{\text{if}} + (N_t^{\text{if}} - N_t^{\text{bulk}})e^{-d/L_d}$ at each face, and $\simeq 2N_t^{\text{if}} - N_t^{\text{bulk}}$ for $d \ll L_d$ — up to a factor-of-two overshoot of the configured interface density. A single-face profile (trap_edge set to left or right) already peaks at exactly N_t^{if} and is not renormalized.

7.10 Band-Gap Grading

Compositionally graded absorbers (e.g. Ga-graded CIGS) are modelled with a *simplified subset* of the SCAPS material-interpolation framework of Burgelman and Marlein: a graded layer is a mixture of two endpoint materials A (front face) and B (back face) with a composition profile $y(x)$ that may be linear, parabolic, or an exponential notch. SCAPS itself admits a wider family — uniform, linear, logarithmic, parabolic, power-law, effective-medium, tabulated and Beta laws, selectable *per material parameter*, with the effective densities of states and the absorption coefficient gradeable alongside E_g and χ . Only the common subset should be compared across the two tools. The graded gap includes an optional bowing term,

$$E_g(y) = (1 - y) E_g^A + y E_g^B - b y (1 - y), \quad (7.35)$$

while the electron affinity interpolates linearly (Vegard's rule). The statistical quantities are anchored to the front face,

$$n_i^2(x) = n_{i,\text{front}}^2 \exp \left[-\frac{E_g(x) - E_g^{\text{front}}}{V_T} \right],$$

which is the ratio form of $n_i^2 = N_C N_V e^{-E_g/k_B T}$ under the assumption that $N_C N_V$ is *constant across the graded layer* — SolarLab does not grade the effective densities of states, so within this model the prefactors cancel exactly and no per-node DOS data are required. The cancellation is a consequence of that modelling choice, not a general identity: it would not hold against a tool that grades N_C and N_V with composition. The SRH references n_1, p_1 are graded so that $n_1 p_1 = n_i^2(x)$ holds node by node. Independent grading of χ and E_g controls the conduction- and valence-band profiles separately, which is how back-surface fields are configured. The transform is applied once at build time; a graded layer automatically refines its mesh so that a steep notch is resolved over several cells. Documented limitation: the *optical* constants are not graded — a graded absorber's absorption edge does not shift spatially, so J_{sc} reflects the electrical effect of grading under uniform optics only.

Chapter 8

Boundary And Initial Conditions

8.1 Potential

The potential is Dirichlet at both contacts:

$$\phi(0) = \phi_{\text{left}}, \quad \phi(L) = \phi_{\text{left}} + V_{\text{bi}} - \text{sgn}(V_{\text{bi}})V_{\text{app}}.$$

Forward bias reduces the built-in field for either stack orientation. The signed V_{bi} comes from the mode selected in the device definition: the difference of two explicit metal work functions, or two semiconductor work functions calculated from the contact-face bands, DOS, doping and active temperature. The latter is a non-degenerate Maxwell-Boltzmann construction; it does not include Fermi-Dirac degeneracy, metal/semiconductor interface dipoles, Fermi-level pinning, or deep-defect charge in the contact-adjacent layer.

The contact-potential source and carrier boundary are configured separately. Changing from an ohmic density pin to a Robin contact changes the exchange kinetics but not V_{bi} . The only exception is an unmodified old YAML containing V_{bi} with no explicit mode: that file retains the historical `flat_band_contacts` coupling so its published benchmark trajectory is not silently moved.

8.2 Carrier Contacts

By default, carrier densities at contacts are ohmic pins derived from local charge neutrality and mass action:

$$np = n_i^2, \quad n - p = N_D - N_A.$$

The implementation evaluates the numerically stable two-branch closed form,

$$n_{\text{eq}} = \eta + \sqrt{\eta^2 + n_i^2}, \quad p_{\text{eq}} = \frac{n_i^2}{n_{\text{eq}}} \quad \left(\eta = \frac{1}{2}(N_D - N_A) \geq 0 \right),$$

and the mirrored branch for p-type material, which avoids catastrophic cancellation at heavy doping. An ohmic (Dirichlet) pin represents an ideal contact with infinite surface exchange velocity: the contact acts as an inexhaustible carrier reservoir at the dark doping equilibrium.

In full mode, selected contacts can instead use a finite-kinetics Robin flux boundary condition:

$$J = \pm qS(u - u_{\text{eq}}),$$

where u is n or p , u_{eq} its contact equilibrium value, and S a surface recombination/extraction velocity. $S = 0$ is a perfectly blocking (Neumann) contact, finite S lets the boundary density evolve with a relaxation time $\sim \Delta x/S$, and $S \rightarrow \infty$ recovers the ohmic limit. Selective contacts are modelled by choosing a large S for the extracted carrier and a small S for the blocked carrier. For Schottky-type contacts the equilibrium reference can be taken as the thermionic reservoir density at the barrier, $n_{\text{eq}} = N_C e^{-\phi_B/V_T}$ (and the N_V analogue for holes), which makes the contact carrier supply work-function-limited rather than doping-limited.

8.3 Ion Boundaries

Both positive and negative mobile ions use zero-flux boundaries. This reflects the physical assumption that ions redistribute inside the device but do not leave through the contacts; it also makes the total ion content an exact conserved quantity of the simulation.

8.4 Initial States

The dark equilibrium state starts from charge neutrality and mass action. The illuminated state is obtained by integrating the transient equations under illumination at the starting voltage until the coupled electron/hole/ion system reaches its light-soaked quasi-steady state.

For J-V, impedance, and degradation, this light-soaked initial state matters because ion and carrier memory affect the measurement: the same voltage grid scanned from a different pre-conditioning state produces a different transient current. Within this model that mobile-ion memory *is* the hysteresis mechanism, and the resulting loop is physical rather than a numerical artifact — but only once the time step, mesh, and convergence checks confirm it, since an unconverged transient produces a spurious loop of its own. It should not be read as a claim about experiment: measured J-V hysteresis in perovskite devices can also carry interface trapping, reversible polarization, slow contact processes, ion-electron coupling, measurement capacitance, and pre-conditioning differences, none of which this model separates.

Chapter 9

Physics Tiers

SolarLab uses three physics-activation modes:

Mode	Active physics	Typical use
legacy	disables upgraded hooks	historical benchmark reproduction
fast	build-once upgrades, no expensive per-RHS hooks	screening
full	every configured hook supported by the selected driver	maximum configured feature set

The distinction between `fast` and `full` is architectural: *build-once* physics is folded into the cached material arrays before the time integrator starts, while *per-evaluation* physics must be recomputed from the evolving state inside every right-hand-side call (Chapter 10). The flag matrix is:

Physics hook	legacy	fast	full	Evaluation
Band offsets + thermionic emission	off	on	on	build-once
Transfer-matrix optics	off	on	on	build-once
Dual mobile-ion species	off	on	on	build-once
Spatial trap profiles	off	on	on	build-once
Temperature scaling	off	on	on	build-once
Photon recycling (net coefficient)	off	on	on	build-once
Self-consistent radiative reabsorption	off	off	on	per-evaluation
Field-dependent mobility	off	off	on	per-evaluation
Selective/Robin contacts	off	off	on	per-evaluation

Legacy disables every upgraded hook and is regression-tested against pinned IonMonger benchmark configurations and metrics (a specific configuration, sweep protocol, and metric envelope — see Chapter 16 — not a blanket equation-level reproduction claim across IonMonger’s full parameter space).

One documented exception crosses the tier matrix: the historically named `flat_band_contacts` option activates the finite-kinetics Robin contact path on all four carrier/side channels on *every* tier. With

an explicit `built_in_potential_mode` it does not choose the Poisson voltage; it changes carrier exchange only. The `off/off/on` row for selective contacts describes the explicit user-supplied `S_*` fields.

The tier is a ceiling, not a command to fabricate missing data — a hook activates only when the configuration supplies the parameters it needs. For example:

- TMM requires `optical_material`.
- Field-dependent mobility requires nonzero `v_sat_*` or `pf_gamma_*`.
- Selective contacts require finite `S_*` values.
- Radiative reabsorption requires TMM/photon-recycling support.

This “enable-if-configured” pattern makes the tier strings safe to set on any preset: flags whose prerequisites are absent stay silent instead of forcing physics the stack cannot support.

The physical QF interface boundary is not a fourth tier. Select it with `solver=quasi_fermi` and `interface_boundary=true`. It supports fewer optional models than the `full` tier. If the request includes an unsupported full-tier hook, SolarLab raises an error before solving.

Chapter 10

Numerical Method

driver	unknowns	discrete model	solve and certificate
transient	$[n, p, P^+, (P^-)]$	SG carrier/ion continuity Poisson at each RHS call	Radau IIA history-preserving state
steady_state	$[\ln n, \ln p]$	1D density RHS fixed ion profile	direct Newton DC residual checks
quasi_fermi	$[\Delta\phi_n, \Delta\phi_p]$	eliminated Poisson response optional local interface solve	Newton continuation cell, current, Poisson certificates
QF frequency	complex small signal	linearization about a certified QF operating point	refined complex solve all-face admittance continuity
2D	$[n(y, x), p(y, x)]$	2D SG + sparse Poisson static ion background	Radau-LU lateral/vertical diagnostics

Driver choice fixes the unknowns, supported physics,
and required certification checks.

Figure 10.1: Numerical drivers, variable sets, and certification paths

The drivers share device construction, grid utilities, Poisson operators and transport definitions, but they do not solve the same algebraic problem. The transient driver advances density and configured ion states in time. The direct steady-state and QF drivers solve different DC variable sets, while the QF frequency path linearizes about a certified operating point. The 2D driver uses a separate sparse Poisson operator and holds the configured ion charge fixed during a J-V run. A J-V point is reported only after the selected driver has completed its own relaxation or residual checks.

10.1 Spatial Grid

The 1D solver builds a multilayer grid over the electrical layers only; substrate layers are excluded from the electrical grid but retained for TMM optics. Within each layer of thickness L the $N + 1$ nodes follow a hyperbolic-tangent stretching,

$$x(\xi) = \frac{L}{2} \left[1 + \frac{\tanh(a \xi)}{\tanh a} \right], \quad \xi \in [-1, 1], \quad a = 3, \quad (10.1)$$

which clusters nodes toward both layer boundaries. This resolves the space-charge regions and interface carrier gradients — where densities vary over Debye-length scales — without wasting nodes in the quasi-neutral interior. Layer grids are concatenated with duplicate interface points removed, and a compositionally graded layer is automatically refined by an integer multiplier so a steep band-gap notch spans several cells.

An optional top-level `electrical_grid` block makes the allocation explicit. Its `interval_weights` and `alphas` maps are keyed by electrical-layer name and must cover those layers exactly. Positive interval weights divide the global `N_grid` budget by deterministic largest-remainder allocation, with at least one interval in each layer and exactly `N_grid` intervals overall. Each positive `alpha` then sets that layer's `tanh` clustering strength. With no block, the historical equal-per-layer grid and grading multiplier are unchanged. A custom weighted grid cannot be combined with `grading_N_mult > 1`, because the two mechanisms would otherwise compete for the same interval budget.

10.2 Method Of Lines And State Vector

The PDEs are discretized in space first, producing a coupled system of ordinary differential equations in time — the method-of-lines approach. The packed state vector contains the nodal values

$$\mathbf{y} = (n, p, P [, P^-] [, \mathbf{s}_{if}]),$$

with the optional negative-ion block in dual-ion mode and an optional trailing block of interface-plane carrier states used by the steady-state SCAPS-comparison machinery. The electrostatic potential is *not* a dynamical unknown: Eq. 7.1 is elliptic and is re-solved exactly, from the instantaneous charge configuration, inside every right-hand-side (RHS) evaluation. This eliminates any splitting error between electrostatics and transport at the cost of one linear solve per evaluation, which the prefactored Poisson path below makes essentially free.

The exclusive physical interface boundary follows a different bookkeeping rule. Its four plane densities per interface are solved and eliminated inside the residual evaluation, so they do not enlarge the packed device state. This is an algebraic constitutive closure nested inside the quasi-Fermi solve, not a fast transient appended to the method-of-lines vector.

10.3 Scharfetter-Gummel Flux Discretization

Central differencing of the drift-diffusion current (Eq. 7.6) becomes unstable when the potential drop per cell exceeds a few V_T — the standard failure of naive discretizations in depletion regions. The Scharfetter-Gummel (SG) scheme instead solves the constitutive relation *exactly* on each cell under

the assumptions of constant current and constant field between adjacent nodes; the resulting face flux is the optimal exponential fit,

$$J_{n,i+1/2} = \frac{qD_n}{\Delta x} [B(\xi) n_{i+1} - B(-\xi) n_i], \quad \xi = \frac{\phi_{n,i+1} - \phi_{n,i}}{V_T}, \quad (10.2)$$

$$J_{p,i+1/2} = \frac{qD_p}{\Delta x} [B(\xi) p_i - B(-\xi) p_{i+1}], \quad \xi = \frac{\phi_{p,i+1} - \phi_{p,i}}{V_T}, \quad (10.3)$$

with the Bernoulli function

$$B(\xi) = \frac{\xi}{e^\xi - 1}. \quad (10.4)$$

The scheme reduces to central differencing in the diffusion limit ($|\xi| \ll 1$) and to pure upwinding in the drift limit ($|\xi| \gg 1$), remaining stable at arbitrarily large cell Peclet numbers (Scharfetter and Gummel 1969; Selberherr 1984). Its positivity property is a statement about the *spatial semi-discretization* — the face flux is monotone, so the semi-discrete operator does not manufacture negative densities — and does not carry over automatically to the fully discrete transient: the state vector (n, p, P) is advanced by a general-purpose implicit integrator, which does not enforce non-negativity at each internal stage or accepted step, and SolarLab applies no clipping or projection to n and p during the transient. Positivity is enforced by construction in the direct steady-state driver, which solves in logarithmic unknowns (Chapter 10). The scheme rests on three implementation details:

- the drift argument ξ is built from the *band-effective* potentials ϕ_n, ϕ_p of Eqs. 7.8–7.9, so heterojunction offsets and density-of-states contrasts enter the flux without any special-case interface treatment;
- $B(\xi)$ is evaluated in three numerically safe branches: a Taylor series for $|\xi| < 10^{-8}$ (removing the 0/0 singularity), the analytic zero limit for $\xi > 700$ (preventing floating-point overflow of e^ξ), and the `expm1` form elsewhere (preserving precision near zero);
- face diffusivities use the harmonic mean of the adjacent nodal values, the composition consistent with flux continuity across a material discontinuity (it also preserves exact zeros, so an ion-blocking layer boundary carries strictly zero ionic flux).

The divergence of the face fluxes is taken over dual-grid (finite-volume) cells, so the semi-discrete continuity equations conserve charge exactly on the non-uniform mesh. The same generalized SG form discretizes Eq. 7.14, with the electrostatic drop and the change in excess chemical potential folded into the Bernoulli argument. The legacy path instead folds the whole-flux factor of Eq. 7.18 into the face diffusivity.

10.4 Poisson Fast Path

The finite-volume Poisson operator is tridiagonal. Its LU factorization depends only on the grid and the permittivity profile — both constant for the lifetime of an experiment — so it is computed once (LAPACK `dgttrf`) and cached; every subsequent RHS evaluation performs a single $O(N)$ back-substitution (`dgttrs`) with the Dirichlet boundary values absorbed into the right-hand side. This is roughly forty times faster than a general sparse solve and is the reason the per-experiment material cache must be built *outside* the time-step loop.

The same build-once strategy applies to all static coefficient fields: per-node and per-face material arrays, band-edge and density-of-states folds, trap-profile lifetime maps, grading profiles, tunnelling-enhanced Richardson constants, and the TMM generation profile are all assembled once into an immutable cache. Only three hooks are inherently state-dependent and re-evaluated per RHS call — field-dependent mobility, self-consistent radiative reabsorption, and Robin contact fluxes — which is exactly the set excluded from the fast tier.

10.5 Right-Hand-Side Assembly

Each RHS evaluation proceeds in a fixed order:

1. apply contact conditions to the carrier arrays (ohmic Dirichlet pins, or free boundary densities on Robin sides);
2. assemble the space charge ρ (Eq. 7.2);
3. solve Poisson for ϕ via the prefactored path;
4. form the generation profile $G(x)$ (cached TMM or Beer-Lambert; plus the reabsorption source when self-consistent photon recycling is active);
5. recompute face diffusivities from the face field if field-dependent mobility is active;
6. evaluate SG fluxes, apply the thermionic-emission cap at flagged heterointerface faces, and take flux divergences with $G - R$;
7. subtract interface recombination at interface nodes. In the opt-in physical QF boundary, locally eliminate the plane states, zero the replaced SG face, and insert only the conservative reservoir-to-plane drains instead;
8. evaluate ion continuity (and the negative species in dual-ion mode);
9. enforce the boundary treatment (zero time-derivative on Dirichlet-pinned entries; Robin fluxes otherwise), and verify all derivatives are finite.

10.6 Time Integration And Robustness

The semi-discrete system is stiff: electronic relaxation (nanoseconds), dielectric relaxation, and ionic migration (milliseconds-seconds) coexist in one state vector. SolarLab integrates it with SciPy's Radau IIA method — a fifth-order, L-stable, fully implicit Runge-Kutta scheme appropriate for stiff differential systems (Hairer and Wanner 1996) — with BDF as a fallback. Stability of the implicit iteration, not accuracy, sets the practical step size, and a layered set of safeguards keeps difficult J-V steps bounded:

- **Step-size ceiling near flat band.** Close to $V_{\text{app}} \approx |V_{\text{bi}}|$ the Jacobian is nearly singular and Radau's local-error estimator can under-report the truncation error, accepting a single oversized step onto a spurious branch of the implicit equations (visible as an isolated non-physical J-V spike). Every voltage step therefore caps the internal step at $1/20$ of the settle interval.
- **Evaluation budget.** A per-step ceiling on RHS evaluations converts a non-contracting implicit iteration (which would otherwise spin without wall-time bound) into a recoverable failure.
- **Bisection in time.** A failed settle interval is halved and re-chained, up to 2^{10} sub-intervals.
- **Frozen-source retry.** The self-consistent reabsorption source couples all absorber nodes through a non-local integral; at the diode-injection knee this can prevent Newton contraction inside Radau. The failed step is retried with the reabsorption source frozen at its entry-state value — a lag bounded by one settle interval, well below the regression tolerance.
- **Method fallback.** A step that exhausts bisection is attempted once with BDF, which occasionally converges where Radau's error estimator stalls.

- **Finite-value guard.** Any NaN/Inf in the assembled derivatives raises immediately rather than letting the integrator accept corrupted state.

10.7 Direct Steady-State Solver

The transient integrator is the reference engine for ion-coupled dynamics, but steady-state comparisons (e.g. against ion-free steady-state simulators such as SCAPS) and artifact-free open-circuit voltages benefit from solving $F(\mathbf{y}) = 0$ directly on the *same* discretized RHS — one physics implementation, two drivers. The steady-state driver freezes the ion profile from its seed state and applies a damped Newton method with:

- **Logarithmic unknowns** $z = (\ln n, \ln p)$, which enforce positivity by construction and equidistribute sensitivity across densities spanning twenty orders of magnitude;
- **Peak-density residual scaling:** residuals are normalized by the global peak density rather than the local one, because at dark depletion nodes the local-relative rate floors at double-precision cancellation noise and no tolerance could ever be met; on the peak-density scale that noise sits near 10^{-10} while a physical imbalance registers at order unity, and the tolerance translates into a terminal-current error bound of $\sim 10^{-8} A m^{-2}$;
- **Finite-difference Jacobian with chord reuse:** the dense Jacobian is built by forward differences and its LU factorization is reused across chord iterations, rebuilt only when a stale factor loses contraction — the dominant cost saving on warm-started continuation solves;
- **Backtracking line search** on the residual 2-norm, with the Newton step capped in log space;
- **Smoothed thermionic-emission cap:** the hard magnitude-minimum of the TE cap is non-differentiable exactly where it binds; the steady-state driver (and only it) replaces it with a narrow sigmoid blend so Newton retains a usable derivative;
- **Quasi-Fermi-preserving Poisson relaxation:** before Newton starts (and after every transient assist), a nonlinear Poisson solve with Boltzmann-responding carriers at frozen quasi-Fermi levels performs the dielectric relaxation analytically. This step is unconditionally well-posed and removes the seconds-slow relaxation mode that otherwise dominates the Jacobian conditioning in near-insulating layers;
- **Certified fallbacks:** where the TE-cap kink makes Newton stall at an already-physically-converged state, the best iterate is accepted under an explicit residual bound; where Newton cannot finish at all, escalating Radau settles compute the point and are *certified* against the same residual scale; an uncertifiable point raises an error rather than returning an unconverged result.

Steady-state J-V curves are computed by voltage continuation with warm-started seeds and automatic voltage-step bisection; the open-circuit voltage can additionally be located by direct bisection on the steady-state $J(V)$ zero crossing, eliminating sweep-grid interpolation error. Measured agreement between the two drivers on the reference configuration is within 5 mV in V_{oc} and 1% in J_{sc} . Two deep-tail regimes are handled by the transient engine rather than the Newton driver: near-insulating contact doping produces no $J = 0$ crossing below the scan ceiling (a physical absence the transient confirms), and deeply collapsed conduction-band-offset junctions are pathological for any algebraic Newton method and are handled by the transient engine only.

10.8 Cancellation-Safe Quasi-Fermi Steady State

The `quasi_fermi` driver targets steady states in which the terminal current is the small difference between very large forward and reverse SG terms. This is the relevant numerical regime in a highly

doped crystalline-silicon emitter and in the zero-thickness interface closure. Reaching a small algebraic residual in density variables is not enough there: direct subtraction can lose the current before Newton sees it.

SolarLab instead solves for increments of the electron and hole quasi-Fermi potentials,

$$\varphi_n = V_T \ln n - (\phi + \chi), \quad \varphi_p = V_T \ln p + (\phi + \chi + E_g). \quad (10.5)$$

A transport-balanced reference state is retained separately from the Newton increments. Poisson and the Boltzmann carrier response are eliminated to tight tolerance at every residual evaluation. For a face whose two exponential SG legs are a and b , the current is evaluated from the known quasi-Fermi drop δ using

$$a - b = \begin{cases} -a \expm1(-\delta), & \delta \geq 0, \\ b \expm1(\delta), & \delta < 0, \end{cases}$$

rather than by subtracting the two rounded absolute legs. Reference and increment drops are never collapsed into one absolute potential and then differenced again.

Illuminated states are reached by source continuation from dark equilibrium. The Newton termination test is only a numerical stopping condition. A point is returned only when independent gates also pass for the normalized cell residual, the integrated electron and hole continuity defects, the peak-to-peak all-face current spread, and the Poisson residual. The physical interface path adds the local plane-closure residual to that certificate.

For the physical interface boundary, the final Newton stage uses face-drop coordinates rather than nodal quasi-Fermi values. For each carrier the closing drop is formed as

$$d_{\text{last}} = - \sum_{i < \text{last}} d_i,$$

which pins both contact increments exactly while retaining sub-ULP interior drops. A certified state transferred to another mesh is regridded by conservative overlap of these face drops, with the round-off remainder placed on the closing face. A nodal-coordinate solve may be used to locate the basin, but the target mesh is always re-solved and re-certified in face-drop coordinates before it is reported.

This driver is intentionally narrower than the transient engine. It currently supports illuminated, local, ion-free J-V with ideal carrier pins and local recombination. Mobile or background ions, selective contacts, field-dependent mobility, non-local photon recycling, the older dynamic interface-state block, and thermionic/cross-node interface terms outside the exclusive boundary are rejected. A stack marked `jv_solver_policy: cancellation_safe_qf_required` cannot run a production J-V through another variable set; an explicit diagnostic override is labelled `unvalidated` rather than being treated as equivalent physics.

10.9 Operator Splitting For Slow Ion Dynamics

Long-time experiments (degradation) use an operator-splitting step matched to the two-time-scale structure: the ion subsystem is advanced over a macroscopic interval with the carrier densities frozen (Poisson re-solved inside the ion RHS, so the ions always see a field consistent with the *current ion profile*

and the frozen carriers), and the carriers are then re-equilibrated by a short full-system transient of order one carrier lifetime. Re-solving Poisson makes the potential consistent with the instantaneous charge density; it does not keep the electrons and holes quasi-stationary with respect to the evolving ionic field, so this is a first-order operator splitting rather than a fully self-consistent electron-ion advance. Its error grows with the macro-step and with the band-bending change the ions produce over that step, so a macro-step halving study against the fully coupled solution is the appropriate check before quoting long-time results — most importantly near open circuit and near flat band, where the ionic screening moves the bands the furthest. Snapshot J-V measurements inside degradation freeze the ion profile entirely, so each snapshot reports the instantaneous electronic response of the current ionic configuration.

10.10 Terminal Current Evaluation

The reported current density combines conduction, ionic, and displacement contributions at mesh faces:

$$J = J_n + J_p + J_{\text{ion}} + \varepsilon_0 \varepsilon_r \frac{\partial E}{\partial t}.$$

Transient sweeps read the contact-adjacent face, where the displacement term is part of the physical external-circuit current. Steady-state probes (Suns- V_{oc} , EQE) instead take the *median* across interior faces: charge conservation makes $J(x)$ uniform at steady state, so any face is formally correct, but residual ionic and displacement transients concentrate at the boundary faces and the median is robust to those outliers.

Because a median summarises the face-to-face disagreement rather than removing it, both steady-state probes also report how large that disagreement was,

$$\Delta J_{\text{spread}} = \max_x J(x) - \min_x J(x),$$

evaluated over the interior faces only — the two contact-adjacent faces are excluded because their displacement term is physical external-circuit current and legitimately swings on ionic-rich stacks, so including them would report a real boundary transient as a conservation error. The value is returned as `J_spread_max` on both result objects.

SolarLab reports this value but does not use it as an acceptance test. Although $J(x)$ is uniform at an exact steady state, the raw transient endpoint does not support a fixed ΔJ_{spread} threshold for V_{oc} , J_{sc} or FF. On the TMM presets, the median current is already stable while the spread remains non-monotone in settling time and strongly mesh-dependent.

At the raw endpoint, the anomalous spread is dominated by the local error of Radau's final unbounded step: `run_transient` is called with `max_step=inf` and `t_eval=[t_end]`, so near a fixed point the controller can inflate its last step to the scale of the full settling interval. Re-entering the integrator for one short controlled leg reduced the interior spread by about 390x on `nip_MAPbI3` and 39,700x on `ionmonger_benchmark`, while moving the median by no more than $2.4 \times 10^{-4} \text{ A m}^{-2}$. The terminal electron, hole, ionic and displacement components do not show a cancellation large enough to explain that raw endpoint anomaly. After the controlled leg, any remaining conduction-current spread is paired with the displacement term and can represent a genuinely unequilibrated ionic state.

Use the raw spread as a diagnostic, not as a fixed pass/fail threshold for the photovoltaic metrics. A longer nominal settling interval does not necessarily remove the final-step error. Control the final step,

then refine the mesh until the *extracted metrics* stop moving. Metrics extraction interpolates J_{sc} at $V = 0$. For $J_{sc} \neq 0$, it brackets V_{oc} at the first pair satisfying $J_i > J_{tol}$ and $J_{i+1} \leq J_{tol}$, where $J_{tol} = 10^{-3}|J_{sc}|$; a genuine sign crossing is the common case, while a collapsed-current curve may reach this declared resolution floor without changing sign. FF and PCE are then computed over the resolved operating quadrant. The `voc_bracketed` flag reports whether this metric bracket was found and accepted, not whether a literal positive-to-negative sign change occurred. The hysteresis index compares scan directions as

$$HI = \frac{PCE_{rev} - PCE_{fwd}}{PCE_{rev}}.$$

10.11 Two-Dimensional Extension

The experimental 2D solver reuses the same physics on a tensor-product grid: the tanh-clustered stack axis crossed with a uniform lateral axis. The 2D Poisson operator is a five-point finite-volume stencil with harmonic-mean face permittivities, factorized once with a sparse LU and reused per evaluation; the carrier fluxes are vectorized 2D SG forms with the same band-effective potentials, and the thermionic-emission cap applies on vertical (stack-direction) heterointerface faces. Mobile ions are held as a static Poisson background extruded from the 1D solution — the documented 2D model assumption. Lateral boundaries are periodic or zero-flux. The 2D solver is regression-gated against 1D in the lateral-uniform limit to $|\Delta V_{oc}| \leq 0.1$ mV, relative $\Delta J_{sc} \leq 5 \times 10^{-4}$, and $|\Delta FF| \leq 10^{-3}$.

10.12 Numerical Safety

Important safety mechanisms include:

- finite-value guards in the RHS;
- harmonic face permittivity;
- stable Bernoulli-function branches;
- the thermionic-emission cap at abrupt heterointerfaces;
- step-size ceilings, evaluation budgets, and bisection/fallback logic in every voltage-stepping experiment;
- TMM determinant guard;
- median-current steady-state readout;
- explicit `voc_bracketed` flags when a J-V sweep does not resolve an accepted open-circuit bracket;
- steady-state convergence raises on non-convergence.

For publication work, numerical settings are part of the method. Report the grid size, voltage spacing, settling time or scan rate, tolerances when modified, and whether the run used `legacy`, `fast`, or `full` mode.

Chapter 11

Running SolarLab

11.1 Installation

From perovskite-sim/:

```
pip install -e ".[dev]"
```

For the frontend:

```
cd frontend  
npm install
```

11.2 Backend

From the SolarLab root:

```
uvicorn backend.main:app \  
    --host 127.0.0.1 --port 8000 \  
    --app-dir perovskite-sim --reload
```

Do not run uvicorn main:app from inside backend/; the backend imports are written around --app-dir perovskite-sim.

11.3 Frontend

```
cd perovskite-sim/frontend  
npm run dev
```

Then open:

<http://127.0.0.1:5173>

11.4 Python API

```
from perovskite_sim.models.config_loader import load_device_from_yaml  
from perovskite_sim.experiments.jv_sweep import run_jv_sweep
```

```

stack = load_device_from_yaml("configs/nip_MAPbI3.yaml")
result = run_jv_sweep(stack, N_grid=80, n_points=40, v_rate=1.0)

print(result.metrics_fwd.V_oc)
print(result.metrics_fwd.J_sc)
print(result.metrics_fwd.FF)
print(result.metrics_fwd.PCE)

```

11.5 Worked Example 1: Baseline J-V Sweep

This example runs a standard n-i-p MAPbI3 J-V sweep and reports the principal photovoltaic metrics.

```

cd perovskite-sim
python - <<'PY'
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

stack = load_device_from_yaml("configs/nip_MAPbI3.yaml")
result = run_jv_sweep(stack, N_grid=80, n_points=40, v_rate=1.0)
m = result.metrics_fwd
print(f"Voc = {m.V_oc:.4f} V")
print(f"Jsc = {m.J_sc:.2f} A/m^2")
print(f"FF = {m.FF:.3f}")
print(f"PCE = {100*m.PCE:.2f} %")
print(f"Voc bracketed: {m.voc_bracketed}")
PY

```

Metric interpretation:

- J_{sc} is the current density near zero applied voltage.
- V_{oc} is extracted from the first current-resolution bracket, $J_{tol} = 10^{-3}|J_{sc}|$; on an ordinary curve this is the neighborhood of the terminal-current zero crossing. The bracket must also pass the configured thermodynamic voltage ceiling.
- FF and PCE are meaningful only when voc_bracketed is true.
- Forward and reverse metrics can differ when mobile ions retain scan history.

11.6 Worked Example 2: Absorber Thickness Study

This example varies absorber thickness while leaving the rest of the stack unchanged, allowing optical collection and recombination trends to be compared under controlled assumptions.

```

from dataclasses import replace
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

base = load_device_from_yaml("configs/nip_MAPbI3.yaml")
absorber_index = next(i for i, layer in enumerate(base.layers)
                      if layer.role == "absorber")

for thickness_nm in (200, 400, 700, 1000):

```

```

layers = list(base.layers)
layers[absorber_index] = replace(
    layers[absorber_index],
    thickness=thickness_nm * 1e-9,
)
stack = replace(base, layers=tuple(layers))
result = run_jv_sweep(stack, N_grid=80, n_points=30, v_rate=2.0)
m = result.metrics_fwd
print(thickness_nm, m.V_oc, m.J_sc, m.FF, m.PCE)

```

Expected interpretation:

- Thicker absorbers can absorb more light, but can also increase recombination or transport loss.
- If J_{sc} does not change under Beer-Lambert optics, check whether α and Φ are physically configured.
- If FF collapses, inspect spatial profiles and contact selectivity before concluding that the absorber is intrinsically poor.

11.7 Worked Example 3: Enabling TMM Optics

This example compares scalar Beer-Lambert generation with wavelength-resolved transfer-matrix generation.

```

from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

for config in ("configs/nip_MAPbI3.yaml", "configs/nip_MAPbI3_tmm.yaml"):
    stack = load_device_from_yaml(config)
    result = run_jv_sweep(stack, N_grid=60, n_points=21)
    print(config, result.metrics_fwd.J_sc)

```

Expected interpretation:

- TMM includes interference, reflection, parasitic absorption, and spectral weighting.
- TMM results are only as credible as the `optical_material` data and AM1.5G spectrum used.
- EQE and EL should use TMM-enabled configurations, not Beer-Lambert-only presets.

Chapter 12

Backend API

Configuration endpoints:

Method	Path	Purpose
GET	/api/configs	list shipped and user presets
GET	/api/configs/{name}	load one preset
POST	/api/configs/user	save a user-edited stack
GET	/api/layer-templates	layer template library
GET	/api/optical-materials	available TMM optical keys

Preferred experiment interface:

Method	Path	Purpose
POST	/api/jobs	submit {kind, config_path device, params}
GET	/api/jobs/{id}/events	stream progress/result/error/done events

When the backend is running, /docs shows the interactive FastAPI schema and /openapi.json provides the machine-readable version. The streaming job endpoint accepts four top-level fields:

Field	Use
kind	required experiment name
config_path	preset path or name; required unless an inline device is supplied
device	inline device object; takes precedence over config_path for single-stack jobs
params	job-specific settings; defaults to an empty object

The tandem job is the exception: it requires config_path because its YAML contains more than one subcell. A typical J-V submission is:

```
{  
  "kind": "jv",
```

```

"config_path": "configs/cSi_homojunction.yaml",
"params": {
  "solver": "quasi_fermi",
  "N_grid": 200,
  "n_points": 42,
  "V_max": 0.7
}
}

```

Submission returns `{"status":"ok","job_id":"..."}`. The client then opens the event stream for that ID:

SSE event	Payload
progress	stage, current, total, eta_s, and message
result	final serialized experiment result
error	message describing a worker failure
done	empty terminal marker, emitted after either result or error

For `kind="jv"`, the streaming defaults and compatibility rules are:

Parameter	Job default	Use
N_grid	60	global electrical-grid budget
n_points	30	requested voltage samples
v_rate	1.0	transient scan rate; ignored by zero-scan-rate drivers
V_max	driver default	upper requested bias
illuminated	true	quasi_fermi currently rejects dark J-V
solver	transient	transient, steady_state, or quasi_fermi
iface_states	false	older plane-state option for steady_state only
interface_boundary	false	exclusive physical boundary; requires quasi_fermi
interface_transport_model	fermi_richardson	meaningful only with the physical boundary enabled

The synchronous `/api/jv` compatibility route accepts the same named J-V fields but defaults to `N_grid=80` and `n_points=40`. Because `/api/jobs.params` is a generic object, OpenAPI cannot validate every job-specific key before dispatch. Unknown or unused parameters may be ignored; use the table for the selected job and confirm the serialized settings in reproducibility work.

Supported job kinds:

```

jv, current_decomp, spatial, dark_jv, suns_voc, voc_t,
eqe, el, impedance, tpv, degradation, tandem,
mott_schottky, jv_2d, voc_grain_sweep

```

The frontend primarily uses streaming jobs. Long-running experiments report progress through callbacks that the backend converts to SSE frames.

Chapter 13

Experiment Manual

13.1 J-V Sweep

The J-V sweep runs forward and reverse voltage scans while preserving state history. This state-preserving scan is the implemented representation of hysteresis in an ion-coupled perovskite cell.

Outputs:

- V_fwd, J_fwd;
- V_rev, J_rev;
- metrics_fwd, metrics_rev;
- hysteresis_index.

Metrics:

$$FF = \frac{P_{\text{mpp}}}{V_{\text{oc}} J_{\text{sc}}}.$$

$$PCE = \frac{P_{\text{mpp}}}{P_{\text{in}}}, \quad P_{\text{in}} = 1000 \text{ W m}^{-2} \text{ by default (AM1.5G, 1 sun).}$$

The low-level metric extractor accepts an explicit P_{in} . The Python QF J-V driver exposes it as $P_{\text{in_W_m2}}$; the current transient and density-Newton drivers, and all backend J-V routes, do not yet forward it. For those paths, a run that changes illumination intensity or spectrum must recompute the branch metrics from the returned arrays:

```
from perovskite_sim.experiments.jv_sweep import compute_metrics
```

```
m = compute_metrics(result.V_fwd, result.J_fwd, P_in=actual_power_W_m2)
```

Until that recomputation is made, `result.metrics_fwd.PCE` and `result.metrics_rev.PCE` retain the 1-sun denominator of 1000 W m^{-2} . The denominator is fixed by the current driver interface; the optical model does not impose it.

If the sweep never reaches the current-resolution bracket, `voc_bracketed=False`. The same flag is returned when an extracted voltage is rejected by the configured thermodynamic ceiling. In either case V_{oc} , FF , and PCE are sentinel values; inspect the voltage range, band-gap ceiling and point-status records before deciding whether to increase V_{max} .

The J-V job additionally accepts a solver selector: the default transient driver is the state-preserving Radau sweep described above, while `steady_state` routes the same device through the ion-frozen direct Newton driver of Chapter 10, returning a single hysteresis-free steady-state curve. The steady-state driver is the appropriate choice when comparing against ion-free steady-state simulators.

`quasi_fermi` selects the cancellation-safe, certificate-bearing driver. It currently returns an illuminated zero-scan-rate curve only, represented in the common result schema with identical forward and reverse branches and zero hysteresis. `interface_boundary=true` is accepted only with this driver and activates the exclusive zero-thickness physical boundary described in Eq. 7.25. Its `interface_transport_model` is one of `fermi_richardson`, `scaps_thermionic`, or `scaps_thermal_velocity`; choosing a non-default model while the boundary is off is rejected. Dark J-V and ion-history studies remain on the transient path.

The solver choice is not a convenience fallback. A stack whose `jv_solver_policy` requires cancellation-safe quasi-Fermi variables rejects the transient and density-Newton drivers for production use. Conversely, a QF request containing unsupported physics raises an error instead of dropping that physics or silently routing to another solver. Per-point QF status records carry the continuity, current-spread, Poisson, and normalized-residual certificates used to accept the curve.

13.2 Current Decomposition

Current decomposition separates:

- electron current J_n ;
- hole current J_p ;
- ion current J_{ion} ;
- displacement current J_{disp} ;
- total current.

This separation helps identify whether a transient is dominated by electronic, ionic, or capacitive response.

13.3 Spatial Profiles

Spatial-profile runs save snapshots of:

- x ;
- ϕ ;
- E ;
- n ;
- p ;
- P ;
- ρ ;
- applied voltage.

These profiles are essential for interpreting unusual J-V behavior. For example, a high V_{oc} with poor FF may indicate selective-contact or recombination issues rather than poor optical generation.

13.4 Dark J-V

Dark J-V fits the diode equation:

$$J = J_0 \left[\exp \left(\frac{V}{n_{\text{id}} V_T} \right) - 1 \right].$$

The output includes ideality factor n_{id} , saturation current J_0 , and the fitted voltage window.

13.5 Suns- V_{oc}

Suns- V_{oc} varies illumination intensity and extracts V_{oc} at each level. It constructs a pseudo-JV curve:

$$J_{\text{pseudo}}(X) = J_{\text{sc,ref}} - J_{\text{sc}}(X), \quad V_{\text{pseudo}}(X) = V_{\text{oc}}(X).$$

The pseudo-JV representation is less affected by series resistance than a standard terminal J-V sweep.

13.6 $V_{\text{oc}}(T)$

The temperature sweep estimates recombination activation energy by fitting V_{oc} versus temperature:

$$V_{\text{oc}}(T) \approx \frac{E_A}{q} - \frac{AkT}{q} \ln \left(\frac{J_{00}}{J_{\text{sc}}} \right),$$

which follows from the single-diode forms

$$J = J_0 \exp(qV/Ak_B T), \quad J_0 = J_{00} \exp(-E_A/Ak_B T),$$

with ideality factor A .

The intercept at $T = 0$ is reported as E_A in eV. Note that a *temperature-independent* A enters the slope only and leaves that intercept unchanged, so the extracted activation energy is insensitive to the value of A — it is $A(T)$, $J_{\text{sc}}(T)$ and $E_g(T)$ varying across the fitted window that bias it. The linear extrapolation to 0 K is therefore an empirical activation energy for the fitted temperature range, not a zero-temperature physical quantity; report the range, the fit quality, and whether J_{sc} was allowed to vary alongside it.

When comparing $V_{\text{oc}}(T)$ against other simulators, the temperature models are generally not equivalent: SolarLab applies $E_g(T)$, $\mu(T)$, $B_{\text{rad}}(T)$, $n_i(T)$, and Arrhenius ion scaling automatically when temperature scaling is enabled, whereas reference tools (e.g. SCAPS) treat only a subset of quantities as temperature-dependent by default and expect the rest to be supplied per temperature. Align every temperature-dependent parameter explicitly before drawing cross-simulator conclusions.

13.7 EQE / IPCE

EQE is:

$$\text{EQE}(\lambda) = \frac{J(\lambda) - J_{\text{dark}}}{q\Phi_{\text{inc}}(\lambda)}.$$

The incremental current is used *signed* against the solar convention ($J > 0$ at $V = 0$ under illumination), so a negative EQE is reported as such and raises a runtime warning rather than being hidden by an absolute value: a wrong-way incremental current indicates a contact polarity, sign-convention, or dark-baseline convergence problem, and must not be presented as a physical quantum efficiency.

SolarLab computes monochromatic TMM generation, solves the drift-diffusion problem at short circuit, subtracts a dark baseline current J_{dark} (the device settled at $V = 0$ with $G = 0$, which cancels any residual ionic drift that has not fully damped), and integrates against AM1.5G for J_{sc} . The incremental ΔJ definition matches the standard measurement convention. The baseline state is the settled dark short-circuit state; bias-light protocols are not modelled. This experiment requires `optical_material`.

13.8 Electroluminescence

EL uses the Rau reciprocity relation, whose general form is

$$\Phi_{\text{EL}}(\lambda, V) = \text{EQE}_{\text{PV}}(\lambda) \phi_{\text{bb}}(\lambda, T) \left[\exp\left(\frac{qV}{kT}\right) - 1 \right],$$

where EQE_{PV} is the external photovoltaic quantum efficiency (absorption *and* collection). SolarLab evaluates the absorber-absorptance upper bound of this relation,

$$\Phi_{\text{EL}}(\lambda) \approx A_{\text{abs}}(\lambda) \phi_{\text{bb}}(\lambda, T) \exp\left(\frac{qV_{\text{inj}}}{kT}\right),$$

which assumes near-unity carrier collection ($\text{EQE}_{\text{PV}} \approx A_{\text{abs}}$), a flat quasi-Fermi-level splitting equal to qV_{inj} across the absorber, and injection well above thermal ($e^{qV/kT} \gg 1$, so the -1 is dropped). Reciprocity itself further assumes transport that is linear in the excess carrier density and only weakly dependent on the operating point.

The blackbody photon flux is:

$$\phi_{\text{bb}}(\lambda, T) = \frac{2\pi c/\lambda^4}{\exp(hc/\lambda kT) - 1}.$$

The non-radiative voltage loss is:

$$\Delta V_{\text{nr}} = -V_T \ln(\text{EQE}_{\text{EL}}),$$

where EQE_{EL} is the external electroluminescence quantum efficiency (emitted radiative flux over injected carrier flux) evaluated at the injection point corresponding to open circuit. The relation quantifies

the V_{oc} deficit relative to the radiative limit under the reciprocity assumptions above; it is not an unconditional identity at arbitrary injection. This experiment also requires TMM optical data.

13.9 Impedance

Impedance applies a small sinusoidal voltage perturbation around a DC bias, integrates a few AC cycles of the full nonlinear transient at each frequency, and extracts the amplitude and phase of the current response by lock-in demodulation (sine/cosine multiplication followed by low-pass filtering). The displacement current $\varepsilon_0 \varepsilon_r \partial E / \partial t$ is included, so the capacitive branch is physical rather than post-processed. The result is $Z(f)$ in Ωm^2 , suitable for Nyquist and Bode plots.

This general path is a *time-domain, small-amplitude, full-nonlinear-model* method. Because the ionic and electronic subsystems respond on different time scales, it can resolve both relaxations without an equivalent-circuit assumption, provided the selected frequency window spans both time constants and the per-frequency integration reaches a periodic steady state. Too few settling cycles or an excessive perturbation amplitude degrade the extracted spectrum.

The optional method="quasi_fermi_frequency" path is a true frequency-domain linearization about a certified QF DC operating point. Central differences of the conserved carrier storage, rate, face conduction current, and displacement charge form the small-signal operator

$$(i\omega \mathbf{M} - \mathbf{A}) \hat{\mathbf{u}} = (\mathbf{b} - i\omega \mathbf{m}_V) \hat{V}. \quad (13.1)$$

The complex system is row-equilibrated and solved with iterative-refinement diagnostics. Conduction and displacement response are evaluated on every face; the result is returned only if the component-wise backward error and the relative all-face admittance spread pass their gates. The public nominal perturbation must be below 20 mV, although the linear operator itself is solved per volt and is therefore amplitude-independent.

The two methods have different scopes. `transient` remains the ion-aware, nonlinear route. `quasi_fermi_frequency` is restricted to the same local, ion-free model envelope as its DC state and is the certified route used for the crystalline-silicon C-V evidence in Chapter 16. Its addition does not retroactively certify endpoint sampling or periodic settling in the older time-domain path.

13.10 Mott-Schottky

Mott-Schottky wraps dark impedance at a fixed frequency and fits:

$$\frac{1}{C^2} = aV + b.$$

Then:

$$V_{bi}^{app} = -\frac{b}{a} + \frac{2k_B T}{q}, \quad N_{eff}^{app} = -\frac{2}{q\varepsilon_r \varepsilon_0 a}.$$

The `impedance_method` argument selects either impedance engine explicitly. For the certified local silicon case, `quasi_fermi_frequency` carries the QF DC, linear-solve, and all-face current certificates

into every bias point. The default remains transient, which is required when mobile-ion dynamics are part of the intended C-V response.

The $2k_B T/q$ term accounts for the thermally diffuse transition-region edges on both sides of a p–n junction. In the one-sided abrupt-junction approximation, $1/C^2 = 2(V_{bi} - V - 2k_B T/q)/(q\epsilon_s\epsilon_0 N)$, so the bare V -axis intercept $-b/a$ corresponds to $V_{bi} - 2k_B T/q$. The one-edge $k_B T/q$ correction applies to the Schottky form and must not be substituted here. A depletion capacitance has $a < 0$, which makes N_{eff}^{app} positive; the fit does not determine the doping *type*, which must be read off the stack and reported separately from $|N|$.

Both extracted quantities are *apparent* values: the textbook interpretation assumes a one-sided abrupt junction, uniform doping, a known permittivity, a cleanly separable depletion capacitance, and traps/ions that do not respond at the measurement frequency. In thin-film perovskites the geometric capacitance, injection capacitance, interface states, and ionic response can each produce a near-linear $1/C^2$ slope of their own, so for thin fully depleted absorbers the analysis can over-interpret non-depletion response. Report the fit frequency, bias window, and phase alongside the fitted values, and do not report the slopes automatically as physical dopant densities.

Chang’s exact p–n capacitance analysis shows that capacitance integrates the differential mobile-carrier inventory throughout the transition region. Its linear $1/C^2$ intercept can therefore remain below the independently known contact-potential barrier after the $2k_B T/q$ correction. Hufschmidt et al. report a typical barrier-minus-intercept difference of 0.1–0.4 V for silicon p–n junctions. Treat the intercept as V_{bi}^{app} unless a compatible external curve and barrier measurement establish the stronger interpretation.

13.11 Transient Photovoltage

TPV applies a small generation pulse at open circuit. SolarLab enforces the open-circuit condition by adjusting V_{app} so that terminal current remains near zero at each reported time. The voltage decay is fitted as:

$$V(t) \approx V_{oc} + \Delta V_0 \exp(-t/\tau).$$

13.12 Degradation

The degradation experiment is a long-time ion-coupled damage proxy. It advances the ion/carrier system under bias and periodically performs snapshot J-V measurements. During each snapshot, ions are frozen while carriers relax, separating slow ionic redistribution from the instantaneous electronic response.

13.13 Tandem J-V

The tandem driver:

1. runs one combined TMM optical calculation over the full tandem stack;
2. runs independent top and bottom sub-cell J-V sweeps with fixed generation;
3. series-matches at a common current;
4. sums the sub-cell voltages.

This is a SolarLab-native workflow (reference steady-state simulators such as SCAPS handle multi-junction devices by external scripting rather than a built-in driver, so no cross-simulator parity is implied). The current driver neglects luminescent coupling between sub-cells, tunnel-junction resistance, and any electrical inter-cell coupling beyond ideal series current matching.

Some tandem optical constants are documented as placeholder/stub data. These must be replaced before using the tandem driver for material-specific predictions.

13.14 2D J-V

The 2D J-V driver uses a tensor-product grid. It is validated against 1D in the lateral-uniform limit and can add absorber grain boundaries with reduced SRH lifetimes. Ions are frozen as static Poisson background in 2D.

13.15 $V_{oc}(L_g)$ Grain Sweep

The grain sweep repeats 2D J-V simulations over grain sizes L_g and reports V_{oc} , J_{sc} , and FF as functions of grain size. It is intended to quantify microstructure-driven voltage loss within the frozen-ion 2D approximation.

Chapter 14

Shipped Presets

Representative shipped presets:

Preset	Purpose
nip_MAPbI3.yaml	n-i-p MAPbI3, Beer-Lambert optics
nip_MAPbI3_tmm.yaml	n-i-p MAPbI3 with TMM optics
pin_MAPbI3.yaml	p-i-n MAPbI3, Beer-Lambert optics
pin_MAPbI3_tmm.yaml	p-i-n MAPbI3 with TMM optics
ionmonger_benchmark.yaml	Courtier/IonMonger-style benchmark
driftdiffusion_benchmark.yaml	Driftdiffusion-inspired benchmark
cigs_baseline.yaml	CIGS-like inorganic thin-film stack
cSi_homojunction.yaml	crystalline silicon homojunction
radiative_limit.yaml	photon-recycling and radiative-limit checks
selective_contacts_demo.yaml	Robin/selective-contact demonstration
field_mobility_demo.yaml	field-dependent mobility demonstration
tandem_lin2019.yaml	2T tandem workflow demonstration
twod/nip_MAPbI3_uniform.yaml	2D lateral-uniform parity preset
twod/nip_MAPbI3_singleGB.yaml	2D single grain-boundary preset
twod/bcx_combined_demo.yaml	combined 2D advanced-physics demo

Preset values are simulation inputs, not universal material constants. For publication studies, users should document the source of every parameter they change or use.

Chapter 15

Troubleshooting And Diagnostics

15.1 `voc_bracketed=false`

Meaning: the metric extractor did not retain an accepted open-circuit bracket. Usually the sweep stopped before the current reached $J_{\text{tol}} = 10^{-3}|J_{\text{sc}}|$; less commonly, a candidate bracket was rejected by the thermodynamic voltage ceiling. A literal current sign change is not required for a collapsed-current curve.

Likely causes:

- `V_max` is too low;
- the voltage grid is too coarse near open circuit;
- the device is numerically unstable near flat band;
- contact or recombination settings produce an unusual current sign.

Recommended response:

1. Check whether the sampled current ever reaches the declared resolution floor and whether the candidate voltage exceeds the band-gap ceiling.
2. If the scan genuinely stopped short, increase `V_max`.
3. Increase `n_points` or add a finer voltage region near the expected V_{oc} .
4. Inspect point-status records and spatial profiles near the highest voltage.
5. Do not report FF or PCE from an unbracketed result.

15.2 Sentinel Or Zero FF/PCE

Zero FF or zero PCE does not always mean a physically dead solar cell. In SolarLab it can also mean the metric extraction refused to infer a maximum power point because V_{oc} was not bracketed. Always check `voc_bracketed` first.

15.3 TMM Or EQE Fails Because Optical Data Are Missing

EQE and EL require at least one layer with `optical_material`. If a stack only uses scalar `alpha`, it can run Beer-Lambert J-V but not wavelength-resolved experiments. Use a `*_tmm.yaml` preset or add verified n, k data.

15.4 YAML Values Look Numeric But Behave Strangely

YAML 1.1 can parse bare scientific notation such as `1e-9` as a string. SolarLab coerces numeric-looking fields in the config loader, but users should write `1.0e-9` for clarity and to avoid surprises in external tools.

15.5 Slow Or Stiff Runs

Thick absorbers, high fields, sharp interfaces, and ion-coupled transients can make Radau solves slow. Use a coarse grid while developing the configuration, then rerun final cases at publication settings. For CIGS or crystalline silicon, transient J-V may be much slower than equilibrium-style checks.

If a stack marked `cancellation_safe_qf_required` rejects a transient or density-Newton J-V, the message is a capability guard, not a request to loosen tolerances. Select `quasi_fermi` for the certified local model or use the explicit diagnostic override and label the result unvalidated. Likewise, a QF request that contains ions, selective contacts, non-local recycling, or another unsupported hook must be reformulated physically; the driver will not drop the hook on the user's behalf.

15.6 Unphysical Band Barriers

Large or inconsistent jumps in `chi` and `Eg` can create artificial heterojunction barriers. If a device has unexpectedly poor FF or current, plot the band diagram, check the transport-layer band offsets, and verify the contact velocities.

For a physical-interface scan, inspect `numerical_certified` and top-level `certified` separately. The first covers the local state, current, statistics, and mesh protocol. The second also includes enabled external gates. A false top-level value with a true numerical value is a scientifically useful diagnostic, not an externally validated barrier threshold.

15.7 2D Results Differ From 1D

For a lateral-uniform 2D stack, differences from 1D should be small when ions are frozen consistently and the grids are comparable. If the difference is large, check:

- whether 1D ions were allowed to move while 2D ions were frozen;
- lateral boundary condition;
- `Nx`, `Ny_per_layer`, and the settling time (`settle_t`);
- whether a microstructure block was unintentionally active.

15.8 Frontend Job Appears To Hang

Long-running experiments use server-sent events. If the UI does not update, check that the backend was started from the SolarLab root with `--app-dir perovskite-sim`, then verify `/api/jobs/{id}/events` is reachable.

Chapter 16

Validation And Evidence

This edition draws on two source states. Commit fbaef22c records the P1 closure from 2026-08-07. The physical QF interface boundary and explicit contact-potential source were checked in the later working tree on 2026-08-10 and 2026-08-11. The verification table below applies to that working tree. P1 decisions and deferred items remain as recorded in `perovskite-sim/reproducibility/P1_CLOSURE_2026-08-07.md` and `p1_gaps.yaml`.

The later working tree has no commit or tag, and its complete slow lane has not been rerun. This manual therefore describes a development state, not a release. A release record still needs a commit, tag or complete patch digest and a finished slow-lane run for the same source state.

The tables separate three questions. Unit and integration tests check the code path. Regression and mesh tests establish numerical behavior within a stated range. External physical validation requires independent curves or measurements. The reproducibility registry uses `certified`, `partial`, `load_only`, `demo`, and `unvalidated` to mark those differences. P1 closure means that each P1 item either met its stated acceptance criteria or was deferred explicitly. It does not make every model externally predictive or ready for publication.

Gate	Command	Result	Time / record
Current Python non-slow	<code>pytest -m "not slow"</code>	1632 passed, 2 skipped, 256 deselected	152.21 s
Frozen P1 slow lane	<code>pytest -m slow</code>	249 passed, 4 skipped, 1551 deselected	2026-08-07 record
Current interface/QF target	selected unit, integration, backend, and reproducibility tests	115 passed, 4 deselected	2.38 s
Current frontend build	<code>npm run build</code>	passed (TypeScript clean; 86 modules)	1.01 s ^a
Current frontend unit tests	<code>npm run test:run</code>	28 files, 393 tests passed	1.54 s ^a

The passing suites cover:

- core Poisson, recombination, ion, optics, contact, temperature, and trap modules;
- cancellation-safe QF current evaluation, DC certificates, frequency-domain response, physical interface QSS elimination, transport-model gating, and CBO-scan checks;

- J-V, dark J-V, Suns- V_{oc} , EQE, EL, impedance, Mott-Schottky, TPV, degradation, tandem, and 2D workflows;
- backend API and SSE job dispatch;
- frontend result rendering and validation UI;
- IonMonger and Driftdiffusion-inspired benchmark envelopes;
- TMM Jsc baselines;
- photon-recycling voltage boost;
- 1D/2D lateral-uniform parity;
- 2D microstructure regression;
- physical trends for bandgap, thickness, mobility, dark ideality, and Suns- V_{oc} slope.

The current interface/QF target combines the physical interface-plane unit tests, QF steady-state tests, CBO-scan unit/integration tests, continuity, SCAPS loader, backend dispatch, and reproducibility-matrix tests. The frontend checks used an isolated local copy with the exact lockfile because the OneDrive-resident node_modules tree timed out during file reads. No dependency or build output was written back to the repository.

The P1 status is:

Item	Status	Evidence and remaining work
CIGS graded notch	closed	internal orientation, grid, and grading checks passed; external material/device evidence remains partial
SCAPS low-doping ETL	closed	documented contact-model numerical branch, not universal contact validation
Lin 2019 tandem	closed; partial external	reported stack and thicknesses pass the declared window; proxy absorber optics and assumed inputs prevent parameter-free prediction
IonMonger residual steady state	closed	residual-certified frozen-ion short-circuit terminal-current mesh lane only
c-Si QF J-V	closed	certified local QF N=200/300/400 curve; general transient/algebraic driver not implied
c-Si transient/algebraic J-V	deferred to P2	requires a compensated dynamic mass-matrix or equivalent general state
c-Si external C-V parity	deferred to P2	available source is 2D with partial contacts; SolarLab comparison is 1D/full-area and the source deck is unavailable

Item	Status	Evidence and remaining work
Exact external solver curves	deferred to P2	publication-era decks, revisions, preconditioning histories, and arrays are not yet content-addressed

For c-Si QF J-V, all 42 points pass on N=200/300/400. The N=400 metrics are

$$J_{sc} = 356.842862 \text{ A m}^{-2}, \quad V_{oc} = 0.592684 \text{ V}, \quad FF = 0.824938, \quad PCE = 17.4470\%.$$

The N=300-to-400 maximum pointwise difference is 0.8406% of the fine-grid J_{sc} . On the same N=200, $V = 0$ physical state, the QF residual is 5.822×10^{-13} with electron/hole continuity bounds $5.80 \times 10^{-10} / 3.51 \times 10^{-9} \text{ A m}^{-2}$ and face spread $3.41 \times 10^{-10} \text{ A m}^{-2}$. Collapsing that state to absolute densities gives a density-SG residual of 33.08 and an electron continuity bound of 3.18 A m^{-2} . This difference is why the c-Si preset requires the QF driver. It does not establish convergence of the general transient solver.

The QF frequency-domain c-Si lane converges on N=200/300/400 over 10 kHz, 100 kHz, and 1 MHz. At N=400 the 100 kHz Mott-Schottky fit gives an apparent intercept of 0.782066 V and $N_{eff}^{app} = 9.5537 \times 10^{21} \text{ m}^{-3}$; the intercept is 0.111 V below the configured contact potential, within the published distributed-carrier range. The uncalibrated van Nijen comparison is still partial_external_comparison: the local curve is grid-converged but lies 19.7–35.6% above the published 2D Sentaurus values over 0–0.2 V.

The 2026-08-10 physical-interface CBO campaign is a separate working-tree result. With fermi_richardson, fixed contacts, interval weights (1,2,1), alphas (3,3.25,3), and face-drop QF coordinates, the N=40/50/60 one-percent J_{sc} critical intervals are 0.386719–0.387109, 0.382813–0.383203, and 0.380078–0.380469 eV. Their union is 7.031 meV, the midpoint shifts decrease from 3.906 to 2.734 meV, and the reference J_{sc} spread is 1.75×10^{-5} . The finest face-current spread is $1.17 \times 10^{-5} \text{ A m}^{-2}$ and the local QSS residual is 2.53×10^{-15} . These pass the internal numerical gates. The normalized SCAPS J_{sc} error is 0.4744 against a 0.05 gate, so the JSON correctly reports numerical_certified=true but top-level certified=false.

16.1 Numerical Evidence And Model-Scope Figures

The numerical panels below are generated from machine-readable observations, not from values copied into the plotting script. The generator checks the registered grid ladders and evidence status before writing a figure. A missing field or a changed certification state stops generation. Test counts and qualitative assertions remain in native tables because their row labels, rather than a plotted magnitude, carry the information.

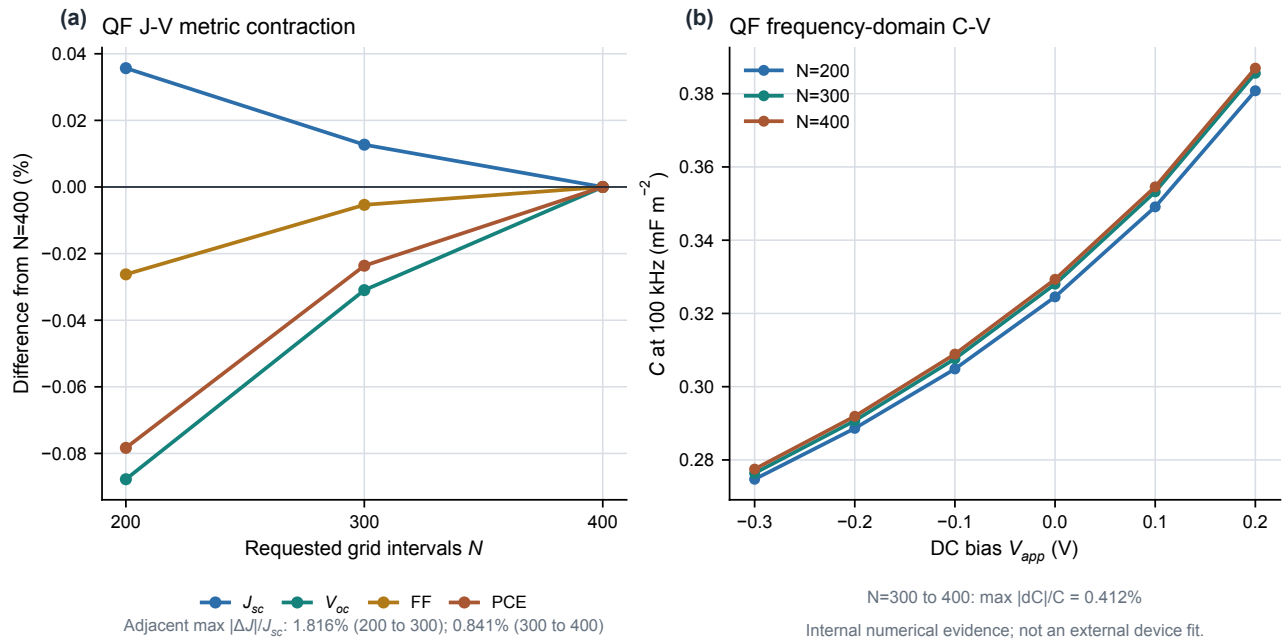


Figure 16.1: Registered c-Si QF J-V and C-V grid-ladder observations

Panel (a) reports each J-V metric relative to the $N=400$ result and gives the maximum pointwise current change between adjacent grids. It is a contraction check against the finest registered grid, not a fitted order-of-accuracy claim against an exact solution. Panel (b) shows the stored 100 kHz capacitance values for the same $N=200/300/400$ ladder. Both panels are internal numerical evidence for the restricted QF model; neither is an external c-Si device fit.

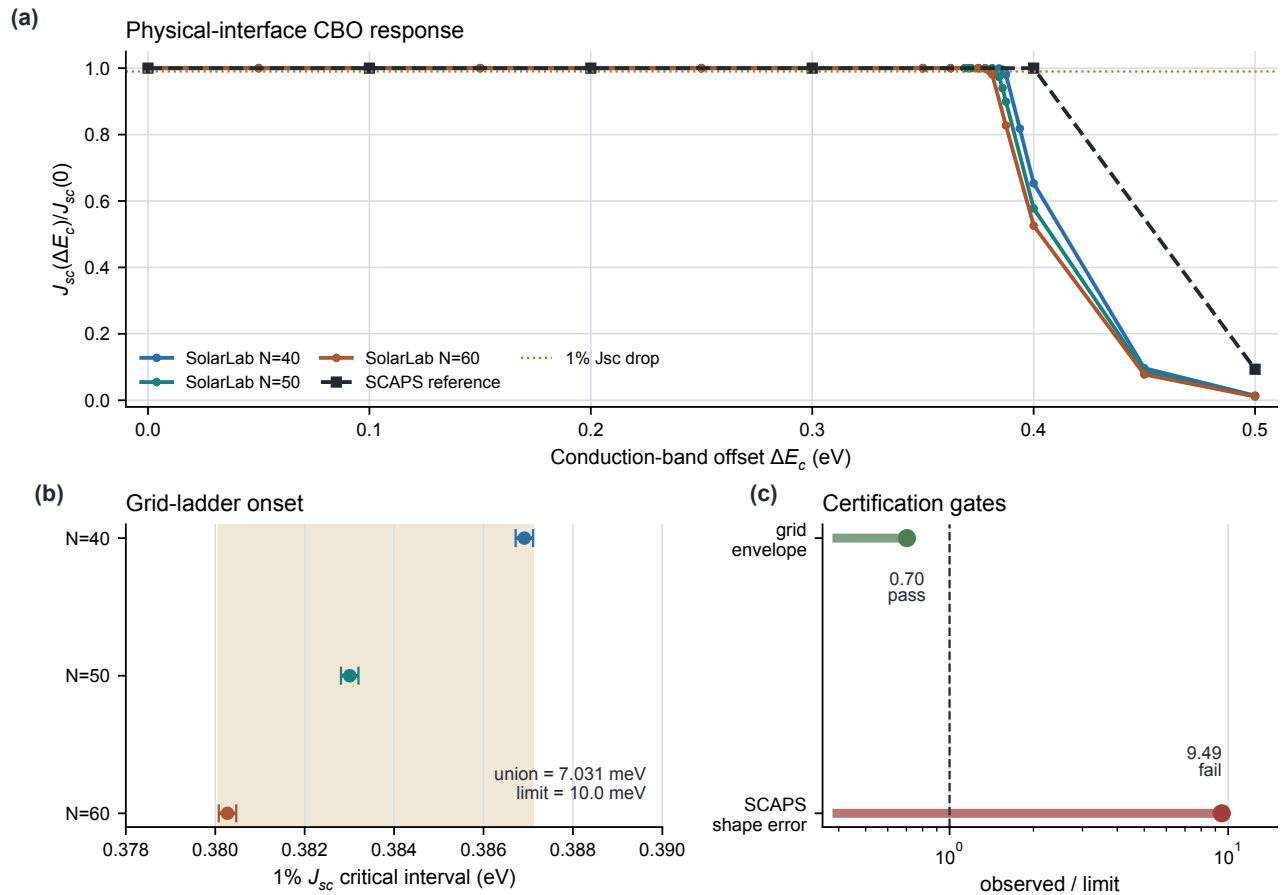
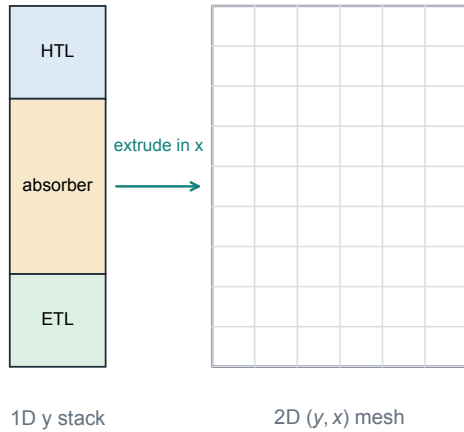


Figure 16.2: Physical-interface CBO response, grid contraction, and certification gates

The CBO figure reads the complete scan JSON. The SolarLab curves use the physical zero-thickness interface with `fermi_richardson`; the dashed curve is the registered SCAPS reference. The $N=40/50/60$ one-percent- J_{sc} intervals form a 7.031 meV union, below the 10 meV internal numerical limit. The external normalized-shape error is 0.4744 against a 0.05 limit. The figure therefore shows both parts of the result: the grid certificate passes, while the top-level external certificate remains false.

(a) Registered lateral-uniform parity domain



nip_MAPbl3_uniform | frozen ions | matched vertical grid
periodic lateral boundary condition

(b) Physics included in the comparison

electrons and holes	active
2D sparse Poisson + band offsets	active
configured ion charge	static background
lateral grain-boundary lifetime field	available
mobile-ion dynamics during 2D J-V	not included
interface SRH / defects / physical QF boundary	not connected

Quantitative 1D/2D parity is meaningful only when both drivers
solve the same configured physics.

Figure 16.3: Registered 1D/2D parity domain and current model scope

The 2D panel is a scope diagram rather than a substitute for stored result arrays. The registered lateral-uniform comparison freezes ions in both drivers, matches the vertical grid and uses an interface-free preset. The 2D solver does not currently consume the 1D interface-SRH, defect or physical-QF-boundary machinery. Quantitative 1D/2D comparison on an interface-active stack is therefore outside the validated domain even when both runs finish.

16.1.1 Other Regression Contracts

The following checks remain useful, but their limits are deterministic acceptance contracts rather than statistical uncertainties. They are listed as a table so a tolerance is not mistaken for an error bar.

Contract	Registered value or interval	Interpretation
IonMonger J-V metrics	$V_{oc} = 1.1912 V$, $J_{sc} = 222.59 A m^{-2}$, $FF = 0.7755$, $PCE = 0.2056$; relative limits 2%, 3%, 3%, 5%	pinned reverse-scan regression at $N_{grid}=40$, $n_{points}=20$, $v_{rate}=5.0$; not an experimental uncertainty
TMM J_{sc}	n-i-p 211.02 and p-i-n 216.62 $A m^{-2}$, each with a $\pm 5 A m^{-2}$ regression tolerance	protects the two optical baselines from implementation drift; the tolerance is not a confidence interval
Photon-recycling boost	40–100 mV	literature-informed sanity window for the recomputed radiative-limit on/off difference; the run value is not persisted as a manual data series

Contract	Registered value or interval	Interpretation
Physical trends	band-gap/current, thickness/voltage, mobility/FF, dark ideality and Suns- V_{oc} slope	qualitative model checks; publication plots require the underlying sweep arrays and protocol metadata
2D grain-boundary penalty	5–100 mV	bounded trend gate on the registered frozen-ion microstructure case, not a universal MAPbI ₃ distribution

The IonMonger values were re-pinned in July 2026 from 1.1932 V, 231.70 $A m^{-2}$, 0.7774 and 0.2149 when generation was made photon-conserving. On that configuration the incident photon flux gives $q\Phi = 224.30 A m^{-2}$, so the superseded J_{sc} exceeded the available photon budget. Generation is now integrated over the solver’s dual cells, which removes that surplus. The acceptance percentages were not relaxed.

16.2 Validation Traceability Matrix

Evidence item	Source test or file	What it checks
Current non-slow Python suite	<code>pytest -m "not slow"</code>	broad current-worktree unit/integration coverage for solver, models, backend, experiments, and reproducibility records
Frozen P1 slow lane	<code>perovskite-sim/reproducibility/P1_CLOSURE_2026-08-07.md</code>	expensive regression baselines and the exact acceptance state at P1 closure
Current interface/QF target	<code>tests/unit/physics/test_interface_plane_physical_qss.py</code> , <code>QF/CBO/backend/reproducibility/companions</code>	local interface balance, reciprocal transport choices, QF certificates, dispatch, and scan criteria
P1 decision registry	<code>perovskite-sim/reproducibility/p1_gaps.yaml</code>	closed/deferred decisions, reproduction commands, numerical gates, and remaining scientific limits
c-Si QF J-V	<code>perovskite-sim/tests/regression/test_csi_quasi_fermi_convergence.py</code>	42-point $N=200/300/400$ grid convergence, photon budget, and per-point physical certificates
c-Si QF frequency response	<code>perovskite-sim/tests/regression/test_csi_frequency_domain_impedance.py</code>	depletion capacitance, storage identity, frequency/step stability, current spread, and linear-solve diagnostics
van Nijen c-Si C-V	<code>perovskite-sim/tests/validation/test_csi_vannijen2025_cv.py</code>	uncalibrated partial external comparison with the 2D/full-area geometry mismatch retained
Physical-interface CBO scan	<code>perovskite-sim/outputs/interface-cbo/scan-fermi-edge-qf-grid-40-50-60.json</code>	internal mesh/statistics certificate and the failed external-shape gate in one result object
IonMonger metric envelope	<code>perovskite-sim/tests/integration/test_voc_benchmark.py</code>	benchmark-scale J-V metrics and hysteresis bounds
TMM J_{sc} baselines	<code>perovskite-sim/tests/regression/test_tmm_baseline.py</code>	optical-stack regression stability

Evidence item	Source test or file	What it checks
Photon-recycling boost	perovskite-sim/tests/regression/test_photon_recycling_voc.py	radiative-limit voltage boost sanity
2D parity	perovskite-sim/tests/regression/test_twod_validation.py	lateral-uniform 2D consistency with 1D
Grain-boundary voltage loss	perovskite-sim/tests/regression/test_twod_microstructure.py	bounded microstructure-induced voltage penalty
Frontend production build	npm run build	TypeScript/Vite production surface
Frontend unit tests	npm run test:run	UI result rendering, validation, progress, and workstation state

^a,The frontend checks were run from an isolated /private/tmp copy using the repository's exact lockfile and the local npm cache. The OneDrive-resident dependency tree timed out during direct reads; the local run changes the filesystem location, not the source or dependency resolution.

The current non-slow Python run reports only np.trapz deprecation warnings in the compatibility and manufactured-solution tests. The separate slow command shown in the table is the frozen P1 record unless a newer completed slow pass is stated explicitly; no unfinished run is counted as evidence.

16.3 SCAPS Comparison: Model Alignment Table

Matching input values does not make SCAPS and SolarLab solve the same problem. They differ in operating mode, contact model, interface treatment and discretisation, and these choices can change the figures of merit directly. The table identifies what must be aligned before comparing a result. Each SolarLab entry was checked against the implementation; the last column gives the required comparison condition.

Item	SCAPS	SolarLab	Condition for a valid comparison
Operating mode	Steady state plus linear small signal; no general transient	Radau/BDF transient, density-based direct steady state, cancellation-safe QF steady state, and a restricted QF frequency-domain response	Select the common quantity explicitly. Use steady_state for the established ion-frozen comparison, or quasi_fermi and quasi_fermi_frequency where that local variable set is certified; a transient curve is not definitionally interchangeable with either

Item	SCAPS	SolarLab	Condition for a valid comparison
Contacts	Metal work function, or a flat-band semiconductor work function computed per temperature	Explicit metal work functions, or a Maxwell-Boltzmann semiconductor work-function difference from χ , E_g , DOS, doping and temperature; legacy manual override retained for frozen benchmarks. Ohmic pins and Robin exchange are selected independently	Use <code>metal_work_function</code> when the electrode values are fixed, or <code>semiconductor_work_function</code> only when the outer layers represent the same semiconductor reservoirs as SCAPS. Align the two work functions, contact equilibrium densities and all four surface velocities separately. A doping sweep moves the semiconductor-derived potential but must not move an explicitly fixed metal-work-function difference
Temperature	$N_C, N_V \propto T^{3/2}$; $v_{th} \propto T^{1/2}$; most other quantities constant	$N_C, N_V \propto T^{3/2}$ and $n_1, p_1 \propto n_i(T)/n_i(300)$; μ , D_{ion} , B_{rad} and E_g carry optional laws; per-layer <code>v_th</code> is imported but has no automatic $T^{1/2}$ law	An interface comparison is presently room-temperature unless the missing thermal-velocity law is supplied and tested explicitly
Interface mesh	Two co-located nodes per interface	Default grid removes the duplicate point; the opt-in QF boundary carries four one-sided zero-thickness plane densities eliminated locally	Matching node count or surface velocity is insufficient. State the topology, electrical-grid weights/alphas, and the interface support used by each code

Item	SCAPS	SolarLab	Condition for a valid comparison
Interface transport	Thermionic-emission boundary condition	Default SG face with an empirical TE cap; alternatively an exclusive physical boundary with <code>fermi_richardson</code> , published-SCAPS Boltzmann, or minimum-thermal-velocity exchange	Compare the actual reciprocal flux law and statistics. <code>scaps_thermionic</code> is the closest equation-level compatibility option, but it remains valid only in its dilute plane-state regime
Interface recombination	Pauwels-Vanhoutte interface states with shared occupancy	Default surface SRH uses cross-carrier sampling and a defect-scoped clamp; the exclusive QF boundary uses signed two-sided capture through one shared occupancy	The defaults are not equivalent. For the new physical path enable <code>interface_boundary</code> , report the transport model, and retain the state/DOS and local-residual certificates; older <code>interface_plane_closure</code> and <code>iface_states</code> results are a separate formulation
Tunnelling	WKB, enabled in DC calculations	Static Padovani-Stratton E_{00} exponential enhancement	Disable on both sides, or validate separately. SolarLab's <code>interface_tunneling</code> is off by default, so the aligned condition holds without action
Grading	Several $y(x)$, $P(y)$ and $P(x)$ laws, including optical grading	Simplified E_g and χ grading; optics are not graded	Compare only the common subset. A graded absorber's absorption edge does not shift spatially in SolarLab, so J_{sc} reflects the electrical effect alone
Mesh	Interface refinement, iterative adaptive	Tanh clustering, optionally with fixed-budget per-layer weights and alphas	Demonstrate mesh convergence of the extracted metrics; equal node counts do not imply equal resolution or allocation

Item	SCAPS	SolarLab	Condition for a valid comparison
Solution method	Gummel with Newton-Raphson sub-steps	Radau/BDF, density-log Newton, or QF Newton with eliminated Poisson and exact face-drop coordinates	Compare accepted residuals, continuity and all-face current conservation, not iteration counts

Two differences dominate the comparison used in this chapter. The optical models are the first. SCAPS integrates a scalar absorption coefficient with no Fresnel reflection at the transport-layer boundary, whereas SolarLab runs a coherent transfer-matrix calculation over the full stack. On the mirrored configuration this accounts for a J_{sc} of 218.2 A m^{-2} against the SCAPS 262.8 A m^{-2} , a -17% difference. This value uses the photon-conserving generation quadrature; point-sampling the front-weighted generation profile would over-count absorbed photons and understate the model difference. Resolving it requires a SCAPS-matched optical calculation; changing an electrical parameter does not address the source of the discrepancy.

The second difference is the set of default choices. Contacts, interface recombination, transport statistics and thermionic normalisation all have alternatives that are off by default. Turning one on does not by itself validate the model. The physical interface boundary now solves transport and shared-occupancy SRH conservatively, but the current CBO campaign passes only the internal grid tests and still fails the declared SCAPS shape test. Reports must state the selected option and whether the agreement is internal numerical or external.

Chapter 17

Limitations And Best Practices

17.1 Model Assumptions

SolarLab is a numerical research simulator. Predictive accuracy for a material stack requires physically justified inputs and a validation envelope that covers the intended regime.

17.2 Known Formulation Limitations

The following model-formulation limitations apply, each flagged at its point of use in Chapter 7:

- the default thermionic-emission bound is an empirical density-weighted cap, not the dimensionally normalized Richardson-Dushman current. The physical emission-velocity form is available through `te_physical_norm` and the previously identified sign, barrier-input and DOS-consistency defects are corrected. It remains opt-in because strongly binding faces still expose conditioning limits in some weak-contact, cold-start and deep-injection regimes. Users who enable it should verify equilibrium zero current, bias continuation, grid convergence and the intended operating window;
- the ionic steric flux now defaults to the diffusion-only modified-PNP form (`ion_steric_diffusion_only`), so the crowding chemical potential modifies diffusion without artificially amplifying field drift. The legacy tier retains the superseded whole-flux multiplier for frozen IonMonger comparisons. The current model is still a single-site lattice gas: it includes site exclusion but not ion-ion correlations, and its near-saturation behaviour is covered by flux-level tests rather than a registered device preset;
- the interface-SRH non-negativity clamp applies only at declared-defect interfaces. The older cross-carrier approximation can otherwise use a cross-side majority-density product as its detailed-balance reference, which is not a local mass-action reference. Defect-free interfaces retain signed generation. Reverse-bias or dark-current studies that depend on the sign of interface generation should use the physically referenced interface-plane closure and report that choice;
- the exclusive QF interface boundary removes the support and detailed-balance ambiguities of that default cross-carrier approximation: ordinary SG transport is removed at the boundary, one shared signed occupancy closes capture and emission, and the four positive plane states are eliminated under an explicit residual certificate. Current evidence is internal: the registered N=40/50/60 CBO scan passes its numerical grid and local-residual gates, but its finest normal-

ized SCAPS J_{sc} error is 0.4744 against a 0.05 external-shape gate. The path is a grid-certified development model; these data do not establish an external CBO threshold;

- the photon-recycling redistribution is spatially uniform, with no spatial or spectral reabsorption kernel;
- the Scharfetter-Gummel positivity property is a property of the *spatial* semi-discretization. The transient advances (n, p, P) with a general-purpose implicit integrator and applies no clipping or projection, so non-negativity is not guaranteed at every internal stage or accepted step. The direct steady-state driver does enforce it, by solving in logarithmic unknowns;
- the slow-ion operator splitting freezes the carriers over the macro-step and re-solves Poisson inside the ion RHS. That makes the field consistent with the instantaneous charge density but does not hold the carriers quasi-stationary against the evolving ionic field, so it is a first-order splitting whose error grows with the macro-step and with the ion-induced band bending;
- per-layer interface thermal velocities can now be supplied directly and are imported by the SCAPS loader, but neither they nor the density-of-states prefactor of the per-side interface-defect trap levels carry an automatic temperature law. Interface formulations using them are therefore room-temperature models unless the missing dependence is supplied and validated explicitly.

High-occupancy tests showed that the old whole-flux steric multiplier produced the wrong trend, so the modified-PNP form is now the default. The normalized thermionic cap remains opt-in because convergence has not been demonstrated over a sufficiently broad set of strongly binding cases. These decisions apply only to the tested regimes; they do not rank the two models in every device class.

17.3 Data And Optical Limits

TMM, EQE, and EL depend on tabulated n, k data. `load_nk` does not silently extrapolate outside the source wavelength range. Placeholder optical constants support workflow testing, but they do not support material-specific publication claims.

17.4 Numerical Limits

If a J-V sweep does not bracket V_{oc} , first distinguish a scan that stopped above the current-resolution floor from a bracket rejected by the thermodynamic ceiling. Increase `V_max` only in the first case, and refine the voltage grid near the expected bracket when its sampling uncertainty matters. Sentinel zero FF or PCE should not be interpreted as physical failure until `voc_bracketed` and the associated limits have been checked.

Mott-Schottky fits can be unreliable for fully depleted thin absorbers.

Thick CIGS and crystalline silicon absorbers can be slow in transient mode.

The shipped crystalline-silicon homojunction requires the cancellation-safe QF driver for production J-V calculations. On the same $N=200$, $V = 0$ one-sun state, the QF residual is 5.82×10^{-13} while the collapsed density-SG residual is 33.08. The general transient/algebraic state remains deferred until a compensated dynamic mass-matrix or equivalent representation independently passes the registered voltage and grid gates. A QF result must not be cited as a converged transient result.

The QF frequency-domain path certifies the restricted local c-Si C-V model; it does not repair the default time-domain path, whose endpoint conduction and interval-averaged displacement sampling returned the whole-wafer geometric capacitance in the audited case. Mobile-ion impedance still requires the time-domain model, together with an explicit settling and amplitude study.

17.5 2D Limits

The 2D solver freezes ions. It is suitable for lateral parity and microstructure studies under that approximation, but not for full dynamic ion migration in two dimensions.

17.6 Reporting Checklist

A reproducible SolarLab study should report:

- commit hash;
- device YAML or full parameter table;
- physics tier;
- grid settings;
- solver driver, variable set, and tolerances;
- interface-boundary flag, transport law, and QF coordinate system when used;
- optical data source;
- validation command results;
- evidence label (certified, partial, load_only, demo, or unvalidated) and the acceptance criteria it refers to;
- whether V_{oc} was bracketed;
- known limitations relevant to the chosen experiment.

17.7 Before Publication

Before publishing a SolarLab figure, verify all of the following:

- the device stack and all changed parameters are reported;
- optical constants and material parameters have provenance;
- the chosen physics tier is justified;
- the grid and solver settings are sufficient for the conclusion;
- validation gates relevant to the claimed regime have passed;
- any internal numerical certificate is kept separate from external physical agreement;
- limitations such as frozen 2D ions, placeholder optical data, or screening smoke settings are stated explicitly.

If any item is missing, the result may still be useful for debugging or hypothesis screening, but it should not be described as a calibrated material-specific prediction.

Chapter 18

SolarScale Screening Bridge

SolarLab can consume readiness-gated material records from SolarScale. This is a workflow layer, not the core solver.

Safe mappings include:

- dielectric constant to `eps_r`;
- electron and hole mobility to `mu_n` and `mu_p` after unit conversion;
- ion diffusion to `D_ion`;
- ion activation energy to `E_a_ion`;
- electron affinity (`electron_affinity_ev`) to the absorber `chi` — with the caution that this parameter directly sets the band alignment at both heterojunctions, so its provenance and uncertainty must accompany any screening result that depends on it.

Device-only unknowns should be swept rather than guessed:

- absorber thickness;
- SRH lifetime;
- trap density;
- interface recombination velocity;
- contact band alignment.

Smoke J-V results in screening workflows validate the process. They are not publication-grade device predictions unless production settings, provenance, and validation are also supplied.

Chapter 19

References

1. D. L. Scharfetter and H. K. Gummel, "Large-signal analysis of a silicon Read diode oscillator," *IEEE Transactions on Electron Devices* **16**, 64–77 (1969). doi:10.1109/T-ED.1969.16566.
2. W. Shockley and W. T. Read, "Statistics of the recombinations of holes and electrons," *Physical Review* **87**, 835–842 (1952). doi:10.1103/PhysRev.87.835.
3. R. N. Hall, "Electron-hole recombination in germanium," *Physical Review* **87**, 387 (1952). doi:10.1103/PhysRev.87.387.
4. N. E. Courtier, J. M. Cave, J. M. Foster, A. B. Walker, and G. Richardson, "IonMonger: a free and fast planar perovskite solar cell simulator with coupled ion vacancy and charge carrier dynamics," *Journal of Computational Electronics* **18**, 1435–1449 (2019). doi:10.1007/s10825-019-01396-2.
5. P. Calado, I. Gelmetti, B. Hilton, M. Azzouzi, J. Nelson, and P. R. F. Barnes, "Driftfusion: an open source code for simulating ordered semiconductor devices with mixed ionic-electronic conducting materials in one dimension," *Journal of Computational Electronics* **21**, 960–991 (2022). doi:10.1007/s10825-021-01827-z.
6. M. Burgelman, P. Nollet, and S. Degraeve, "Modelling polycrystalline semiconductor solar cells," *Thin Solid Films* **361–362**, 527–532 (2000). doi:10.1016/S0040-6090(99)00825-1.
7. U. Rau, "Reciprocity relation between photovoltaic quantum efficiency and electroluminescent emission of solar cells," *Physical Review B* **76**, 085303 (2007). doi:10.1103/PhysRevB.76.085303.
8. L. A. A. Pettersson, L. S. Roman, and O. Inganäs, "Modeling photocurrent action spectra of photovoltaic devices based on organic thin films," *Journal of Applied Physics* **86**, 487–496 (1999). doi:10.1063/1.370757.
9. E. Yablonovitch, "Statistical ray optics," *Journal of the Optical Society of America* **72**, 899–907 (1982). doi:10.1364/JOSA.72.000899.
10. D. M. Caughey and R. E. Thomas, "Carrier mobilities in silicon empirically related to doping and field," *Proceedings of the IEEE* **55**, 2192–2193 (1967). doi:10.1109/PROC.1967.6123.
11. J. Frenkel, "On pre-breakdown phenomena in insulators and electronic semiconductors," *Physical Review* **54**, 647–648 (1938). doi:10.1103/PhysRev.54.647.
12. M. A. Green, "Solar cell fill factors: General graph and empirical expressions," *Solid-State Electronics* **24**, 788–789 (1981). doi:10.1016/0038-1101(81)90062-9.
13. J. Nelson, *The Physics of Solar Cells* (Imperial College Press, London, 2003).
14. P. Würfel, *Physics of Solar Cells: From Principles to New Concepts* (Wiley-VCH, Weinheim, 2005).
15. H. J. Snaith, "Perovskites: The emergence of a new era for low-cost, high-efficiency solar cells," *Journal of Physical Chemistry Letters* **4**, 3623–3630 (2013). doi:10.1021/jz4020162.

16. L. M. Pazos-Outón et al., “Photon recycling in lead iodide perovskite solar cells,” *Science* **351**, 1430–1433 (2016). doi:10.1126/science.aaf1168.
17. S. Selberherr, *Analysis and Simulation of Semiconductor Devices* (Springer, Vienna, 1984).
18. E. Hairer and G. Wanner, *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems*, 2nd ed. (Springer, Berlin, 1996).
19. G. F. Burkhard, E. T. Hoke, and M. D. McGehee, “Accounting for interference, scattering, and electrode absorption to make accurate internal quantum efficiency measurements in organic and other thin solar cells,” *Advanced Materials* **22**, 3293–3297 (2010). doi:10.1002/adma.201000883.
20. F. A. Padovani and R. Stratton, “Field and thermionic-field emission in Schottky barriers,” *Solid-State Electronics* **9**, 695–707 (1966). doi:10.1016/0038-1101(66)90097-9.
21. Y. P. Varshni, “Temperature dependence of the energy gap in semiconductors,” *Physica* **34**, 149–154 (1967). doi:10.1016/0031-8914(67)90062-6.
22. M. S. Kilic, M. Z. Bazant, and A. Ajdari, “Steric effects in the dynamics of electrolytes at large applied voltages. II. Modified Poisson-Nernst-Planck equations,” *Physical Review E* **75**, 021503 (2007). doi:10.1103/PhysRevE.75.021503.
23. H. Pauwels and G. Vanhoutte, “The influence of interface state and energy barriers on the efficiency of heterojunction solar cells,” *Journal of Physics D: Applied Physics* **11**, 649–667 (1978). doi:10.1088/0022-3727/11/5/009.
24. M. Burgelman and J. Marlein, “Analysis of graded band gap solar cells with SCAPS,” *Proceedings of the 23rd European Photovoltaic Solar Energy Conference*, 2151–2155 (2008).
25. Y. F. Chang, “The capacitance of p-n junctions,” *Solid-State Electronics* **10**, 281–287 (1967). doi:10.1016/0038-1101(67)90014-7.
26. M. Hufschmidt et al., “The influence of edge effects on the determination of the doping profile of silicon pad diodes,” arXiv:1605.00778v1 (2016).
27. J. Verschraegen and M. Burgelman, “Numerical modeling of intra-band tunneling for heterojunction solar cells in SCAPS,” *Thin Solid Films* **515**, 6276–6279 (2007).
28. D. A. van Nijen et al., “The nature of silicon PN junction impedance at high frequency,” *Solar Energy Materials and Solar Cells* **282**, 113383 (2025). doi:10.1016/j.solmat.2024.113383.

Chapter 20

Appendix A: Documentation Sources

This manual describes the current implementation, the reason for unusual defaults, known limitations and the evidence behind the stated conclusions. It does not keep a development diary. Include a failed approach only when it changed the physical interpretation or still determines a default or operating limit. Detailed iteration history remains in `perovskite-sim/CLAUDE.md`, the reproducibility records and version-control history.

The manual was drafted from:

- `docs/solarlab_manual_source_dossier.md` (removed 2026-07-29; in git history at `db700fe`)
- `docs/manual/valEvidence260723.md` (historical 2026-07-23 pass)
- `docs/manual/valEvidence260519.md` (superseded 2026-05-19 pass)
- `perovskite-sim/reproducibility/P1_CLOSURE_2026-08-07.md`
- `perovskite-sim/reproducibility/p1_gaps.yaml`
- `perovskite-sim/docs/physical-interface-cbo-scan.md`
- `perovskite-sim/outputs/interface-cbo/skan-fermi-edge-qf-grid-40-50-60.json`
- `perovskite-sim/CLAUDE.md`
- `README.md`
- `perovskite-sim/README.md`
- `perovskite-sim/backend/README.md`
- `perovskite-sim/perovskite_sim/models/`
- `perovskite-sim/perovskite_sim/physics/`
- `perovskite-sim/perovskite_sim/solver/`
- `perovskite-sim/perovskite_sim/experiments/`
- `perovskite-sim/perovskite_sim/experiments/quasi_fermi_steady_state.py`
- `perovskite-sim/perovskite_sim/experiments/quasi_fermi_impedance.py`

- `perovskite-sim/perovskite_sim/physics/interface_plane.py`
- `perovskite-sim/perovskite_sim/twod/`
- `perovskite-sim/backend/main.py`
- `perovskite-sim/frontend/src/types.ts`

This PDF is intended to be defensible as a project manual. For a journal supplement, add parameter provenance for every material value and verify the full bibliography format against the target journal style.

Chapter 21

Appendix B: Figure Sources

Figure	Source	Reproducibility note
Architecture and data flow	docs/manual/generate_manual_figures.py	Implementation diagram checked against the shared schema, material-cache and explicit driver dispatch paths.
Device and contact boundary	perovskite-sim/perovskite_sim/models/device.py, perovskite-sim/perovskite_sim/solver/mol.py	Shows YAML layer order, signed contact-potential sources and the independent carrier-contact law.
Band and interface conventions	perovskite-sim/perovskite_sim/physics/continuity.py, perovskite-sim/perovskite_sim/physics/interface_plane.py	Schematic energy convention; no material-specific solved band profile is implied.
Solver topology	perovskite-sim/perovskite_sim/experiments/ and perovskite-sim/perovskite_sim/twod/	Separates transient, direct steady-state, QF DC, QF frequency and 2D variable sets.
c-Si QF convergence	perovskite-sim/reproducibility/config_benchmark_matrix.yaml	Reads registered N=200/300/400 J-V metrics, adjacent-curve changes and 100 kHz C-V arrays.
Physical-interface CBO	perovskite-sim/outputs/interface-cbo/scan-fermi-edge-qf-grid-40-50-60.json	Reads the full traces, critical intervals and separate internal/external certification gates.
2D model scope	perovskite-sim/perovskite_sim/twod/solver_2d.py, perovskite-sim/tests/regression/test_twod_validation.py	Identifies the registered parity domain and the interface-physics boundary without inventing unrecorded observations.

Chapter 22

Appendix C: Glossary

Term	Meaning
AM1.5G	Standard terrestrial solar spectrum used for photovoltaic testing.
Beer-Lambert	Scalar absorption law using an absorption coefficient α .
Bernoulli function	$B(\xi) = \xi/(e^\xi - 1)$; the weighting function of the Scharfetter-Gummel flux.
Detailed balance	Requirement that every generation/recombination channel vanishes exactly at thermodynamic equilibrium.
Displacement current	Capacitive current $\varepsilon_0 \varepsilon_r \partial E / \partial t$ included in the terminal current.
Drift-diffusion	Continuum transport model combining diffusion from concentration gradients and drift from electric fields.
FF	Fill factor, the ratio between maximum power and $V_{oc} J_{sc}$.
Hysteresis	Difference between forward and reverse J-V scans caused by state memory, often ionic in perovskites.
Quasi-Fermi potential	Electrochemical carrier potential whose spatial drop drives electron or hole current; SolarLab stores reference and increment separately in cancellation-sensitive solves.
Ideality factor	Slope parameter n_{id} of the dark diode characteristic.
Method of lines	Numerical method that discretizes space first and integrates the resulting ODE system in time.
PCE	Power-conversion efficiency.
Photon recycling	Reabsorption of internally emitted photons, reducing the net radiative loss.

Term	Meaning
Quasi-Fermi level	Electrochemical potential of one carrier population out of equilibrium; flat quasi-Fermi levels mean zero current.
Radau IIA	L-stable fully implicit Runge-Kutta method used for the stiff time integration.
Richardson constant	Prefactor A^* of the thermionic-emission flux across a band offset.
SG flux	Scharfetter-Gummel flux discretization, used for stable drift-dominated carrier transport.
SRH	Shockley-Read-Hall trap-assisted recombination.
Steric site density	Finite-site scale P_{lim} entering the mobile-ion crowding chemical potential; it is not a hard transient-state bound.
Thermionic emission	Interface-limited carrier transport over a heterojunction band offset.
TMM	Transfer-matrix method for wavelength-resolved multilayer optics.
J_{sc}	Short-circuit current density.
V_{oc}	Open-circuit voltage.